EEE/EIE Y2 Smart Grid Project

Group: Watt's Up

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June 2026

Abstract

The Smart Grid is a DC microgrid built from five cooperating SMPS modules. It balances generation, storage and load on a shared 10V bus while minimising net grid cost through import, export and supercapacitor charge or discharge. Each converter regulates its own current flow against a local bus measurement, with a central coordinator setting dispatch commands. The system operates from a 12V primary supply onto a 10V bus with a ±0.1V droop band, allowing each module to act autonomously on the bus voltage signal. A Pico W on every module runs the control loop and provides INA219 or SPI ADC telemetry to a user-facing dashboard. A 3D-printed chassis houses the SMPS boards for airflow and mechanical protection.

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Contents

1 Project Objectives & Management 1
1.1 Project Objectives 1
1.2 Project Management System & Temporal Organisation 1
2 System Architecture & Key Design Decisions 4
2.1 Module Roles 4
2.2 Overall Microgrid Topology 5
2.3 Voltage Architecture: 10V Bus, ±0.1V Droop Band and 12V PSU 5
2.4 Coordination Scheme: Symmetric Droop vs Mutex Arbitration 7
2.5 Communication Protocol 8
3 PV SMPS 10
3.1 Design & Simulation 10
3.1.1 Topology Selection 10
3.1.2 MPPT Strategy: Lookup Table vs P&O 10
3.2 Testing & Implementation 12
3.2.1 Bench Noise Treatment 12
3.2.2 P&O Implementation 12
3.2.3 Mode Comparison Support 12
3.3 Evaluation & Optimisation 12
3.3.1 MPPT Mode Comparison 12
3.3.2 Measured SMPS Efficiency 14
4 Supercapacitor SMPS 17
4.1 Design & Simulation 17
4.1.1 Boost Topology and Boot-with-Empty-Capacitor Rationale 17
4.1.2 Operating Window 10.5 to 17.5 V 17
4.1.3 Command-Driven CC: Rationale for the Departure 18
4.2 Testing & Implementation 18
4.2.1 Inner-Loop Feedforward and Bus-Droop Backoff 18
4.3 Evaluation & Optimisation 18
4.3.1 Energy Balance Method 19
4.3.2 Cycle Test State Machine 19
4.3.3 Measured Results 20
4.3.4 Self-Discharge and Hold-Duration Dependence 22
5 Import & Export SMPS 24
5.1 Design & Simulation 24
5.1.1 Topology Choice 24
5.1.2 Droop Setpoints: 9.9 V Import and 10.1 V Export 24
5.2 Testing & Implementation 25
5.2.1 Anti-Windup and Soft-Start Grace 25
5.3 Evaluation & Optimisation 25
5.3.1 Measured Export Efficiency 26
5.3.2 Measured Import Efficiency 26
6 LED Load SMPS 28
6.1 Design & Simulation 28
6.1.1 Topology 28
6.1.2 Control Mode: CC vs Commanded Power 28

6.2 Testing & Implementation ... 28
6.3 Evaluation & Optimisation ... 29
6.3.1 Power-Following Accuracy ... 29
6.3.2 Bus-Side Measurement Gap ... 30
7 Telemetry Filtering ... 31
7.1 Hardware Averaging ... 31
7.2 Exponential Moving Average ... 31
7.3 Three-Second Block Average ... 32
7.4 Evaluation ... 32
8 Software System ... 34
8.1 Architecture ... 34
8.2 Backend and Control Layer ... 34
8.3 Frontend Dashboard ... 36
8.4 Design Trade-offs ... 36
8.5 On-Module Fallback Dashboard [R WE MENTIONING THIS AS /TLM IS REMOVED] ... 37
9 Algorithm Software ... 38
9.1 Coordinator Loop and Economic Objective ... 38
9.2 Algorithm Framework ... 38
9.3 Naive Baseline ... 39
9.4 Smart V1: Forecast-Aware Scheduling and Arbitrage ... 40
9.5 Evaluation ... 42
9.6 Auxiliary Power Overhead ... 43
10 Mechanical Design & Project Polish ... 44
10.1 3D-Modelled SMPS Enclosure ... 44
11 System-Level Testing & Evaluation ... 45
11.1 Test Scenarios ... 45
11.1.1 Per-Module Efficiency Sweeps ... 45
11.1.2 Communication & Integration Tests ... 45
11.2 Performance Metrics ... 45
11.3 Requirements Verification Matrix ... 46
12 Conclusion ... 47
References ... 48

1 Project Objectives & Management

1.1 Project Objectives

The project delivers a five-module DC microgrid that regulates a shared 10V bus across PV generation, supercapacitor storage, grid import, grid export and a programmable LED load. The control system dispatches power between modules to minimise net cost under a time-varying tariff. The prototype is evaluated on demo day, 18 June 2026, against a published price-and-load scenario.

1.2 Project Management System & Temporal Organisation

Coordinating a seven-member hardware and software team across a five-week delivery window required tooling that off-the-shelf platforms could not provide as a single integrated system. Rather than stitching together Notion, Trello, Microsoft Teams and ChatGPT, each of which solves only a fragment of the problem individually, an AI-powered Project Management System (PMS) was coded by our team for the project. Every layer of the frontend, backend, AI pipeline and database schema was purpose built around our needs.

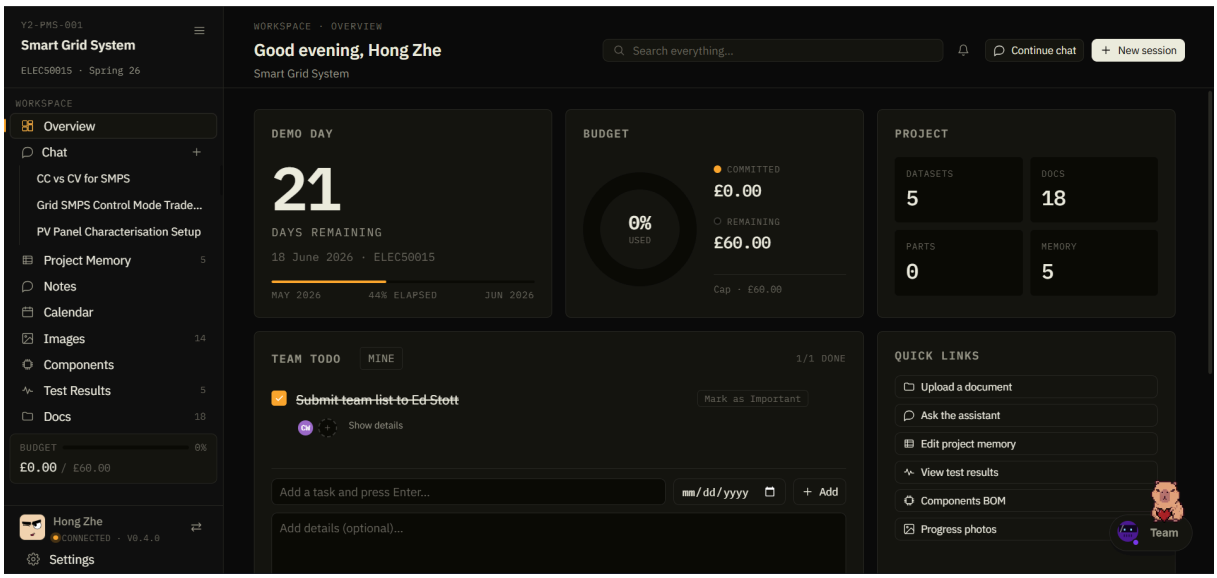


Figure 1.1: PMS dashboard: countdown, budget, stats and todo.

The platform is structured around four modules. A live operational dashboard (Figure 1.1) displays the project's current situation and deadlines. An automated meeting transcript module (Figure 1.2) records, transcribes and structures every meeting into attendee lists, discussion summaries and extracted action items. An AI-driven article and documentation generator named Aria (Figure 1.3) produces project-specific technical writing such as component-selection notes and lab write-ups. A Retrieval-Augmented Generation (RAG) project chat (Figure 1.4) compiles Aria articles, meeting minutes, the GitHub firmware repository, datasheets and test results into context for the AI to analyse and discuss with team members.

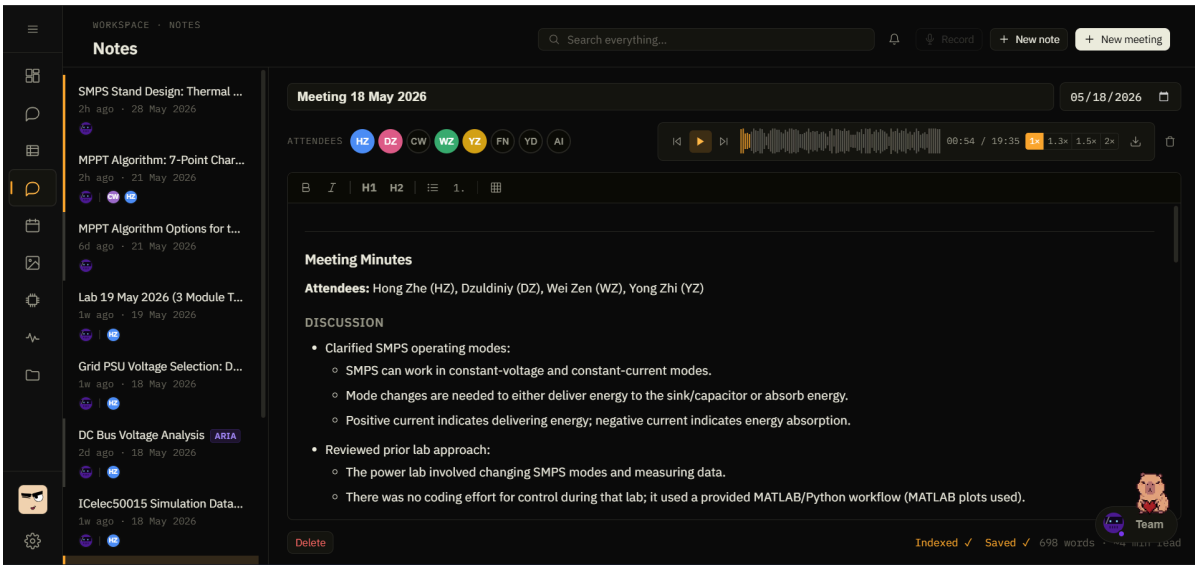


Figure 1.2: Meeting transcript module.

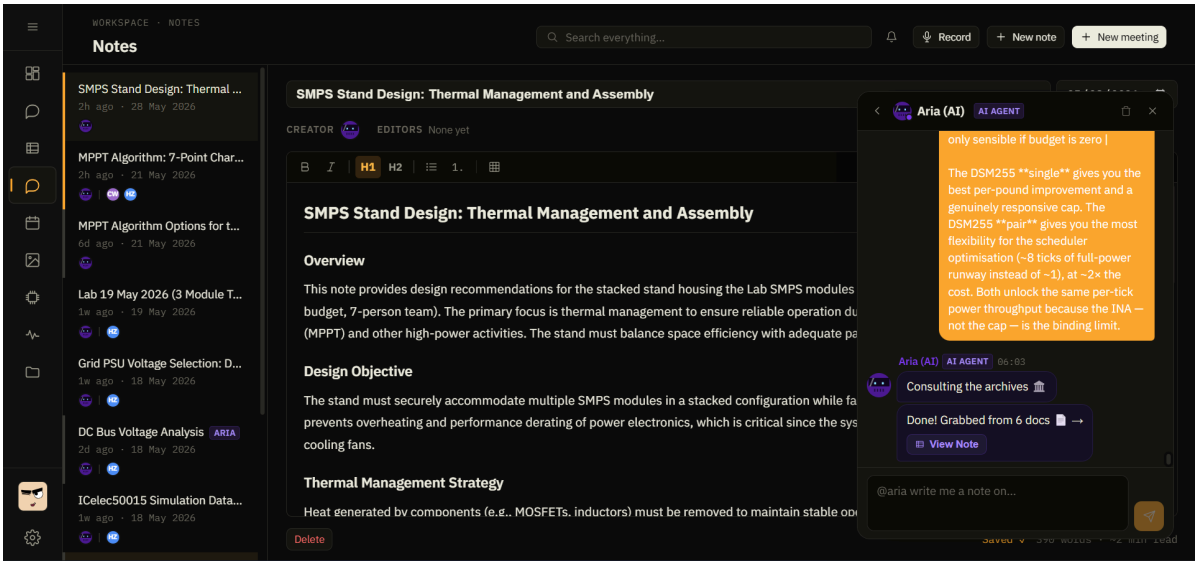


Figure 1.3: Aria article generator.

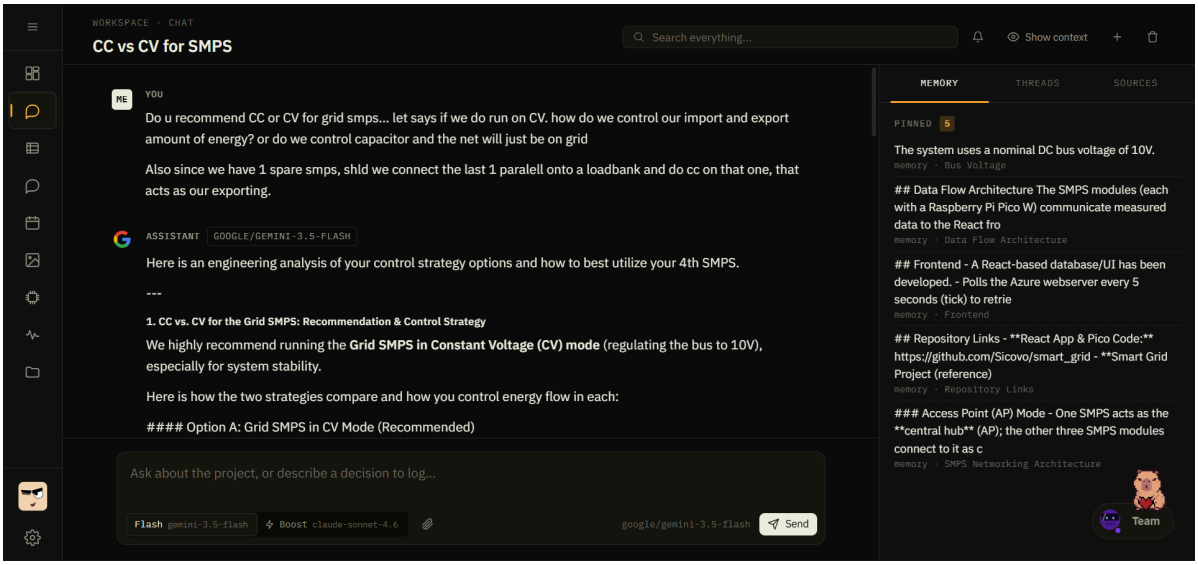


Figure 1.4: RAG project chat with cited sources.

Each feature of the website responds to live changes quickly and related features update accordingly. A transcribed meeting immediately provides the RAG with data, an Aria article updates the dashboard document count, and a new task displays on the calendar and triggers the email reminder service. Temporal organisation is enforced by the countdown and calendar views, which together render the 18 June 2026 demo day as a key deadline to keep in mind. The cumulative effect across the project window has been faster decision-making, fewer repeated discussions, and adherence to the budget.

Several additional features were built other than the four core modules (Figure 1.5). A shared calendar with a Gantt-style timeline consolidates every milestone and submission deadline so that everyone is aware of the project timeline. A project gallery centralises lab photographs that would otherwise be unorganised and scattered across personal phones. A test results module captures raw measurement data with built-in charting, removing the need for data transfer between spreadsheet software. Finally, lightweight personalisation through the team buddy selector motivates team members to actively log on to the platform and increases productivity.

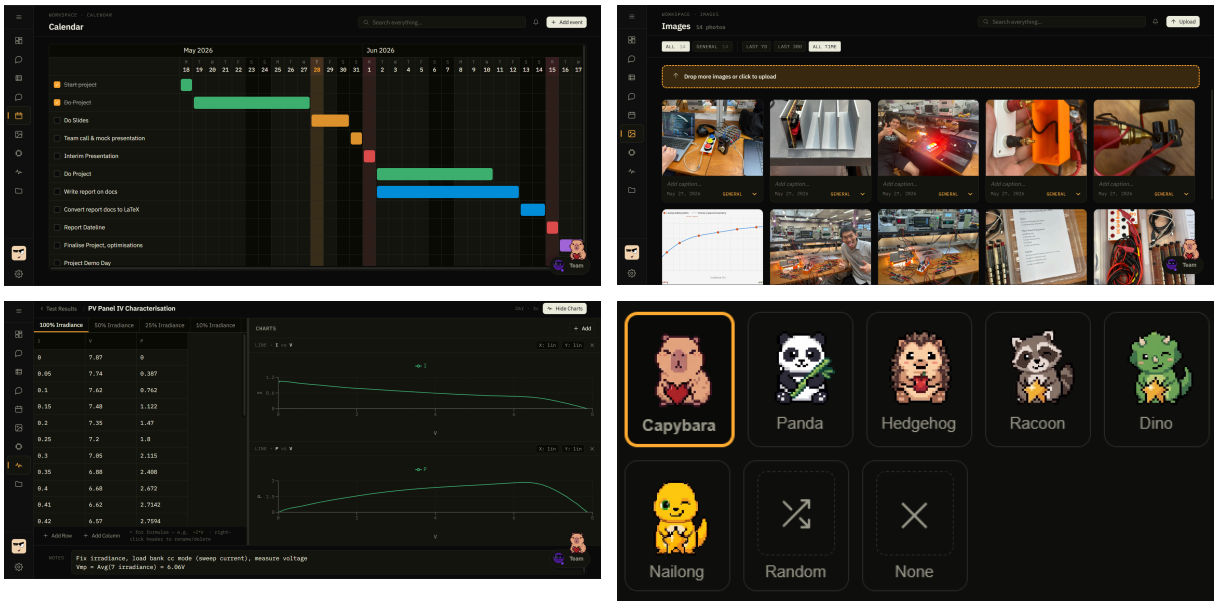


Figure 1.5: Calendar, image gallery, test results, and buddy selector.

2 System Architecture & Key Design Decisions

Architecture decisions set the boundary conditions for every SMPS, control loop and measurement choice. This section covers module roles, the topology that wires them together, and the rationale for the bus voltage, supply voltage, coordination scheme and inter-module communication.

2.1 Module Roles

Module	External component	Topology	Control	Role on the bus
Import SMPS	12V grid PSU	Buck	CV, droop at 9.9V	Sources current when bus $<$ 9.9V
Export SMPS	Sink resistor	Buck	CV, droop at 10.1V	Sinks current when bus $>$ 10.1V
PV SMPS	PV panel	Boost	MPPT	Harvests panel power into bus
Supercapacitor SMPS	10.5 to 17.5V supercapacitor	Bidirectional buck/boost	CC, commanded	Stores or releases on command
LED Load SMPS	3V LED module	Buck (internal)	CC	Drives the LED load

Table 2.1: Per-module topology, external component, control mode and bus role.

The system comprises five SMPS modules connected to a shared 10V DC bus. The two grid-facing modules share a synchronous-buck topology and a common droop, differing only in setpoints; this creates the dead-band discussed in Section 2.4. The Supercapacitor SMPS does not droop and runs as a commanded current source, so the economic algorithm in Section 9 can use it for temporal arbitrage. The LED Load SMPS is internally managed; only its bus-side draw is exposed. All four bidirectional modules use the same power stage in Figure 2.1, with a 100µH inductor and a 100mΩ shunt on Port B. The roles in the table depend on the port connections, not on the hardware, so per-module sections refer back to this single stage.

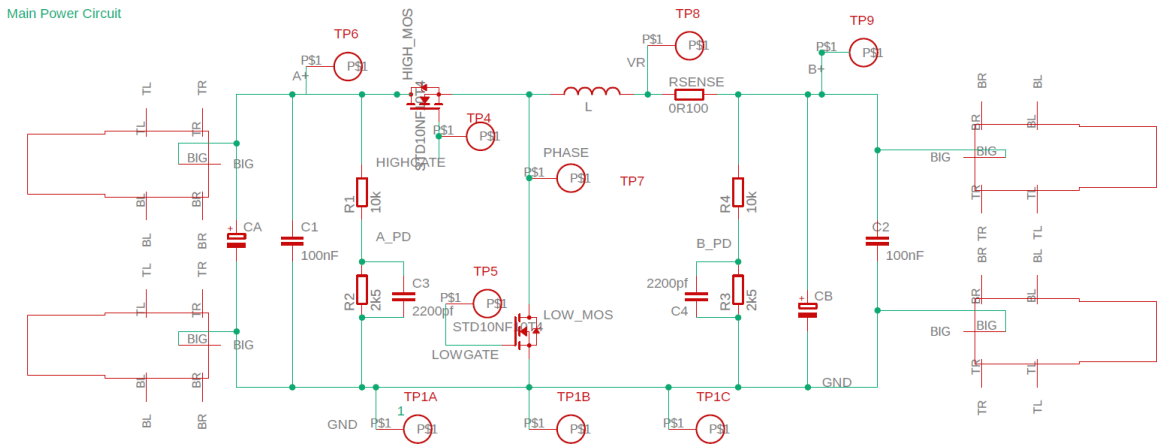


Figure 2.1: Bidirectional SMPS power stage, common to all four modules [1].

2.2 Overall Microgrid Topology

Figure 2.2 shows the complete system. External components sit at the top, each connected through its SMPS module to the shared 10V DC bus. A dedicated Pico W on every module runs the local control loop. A FastAPI backend polls an Azure-hosted web service every 5 seconds for demand, price and irradiance, computes the dispatch decision (Section 9), and pushes setpoints to each module over HTTP. The operator dashboard renders telemetry from the same backend.

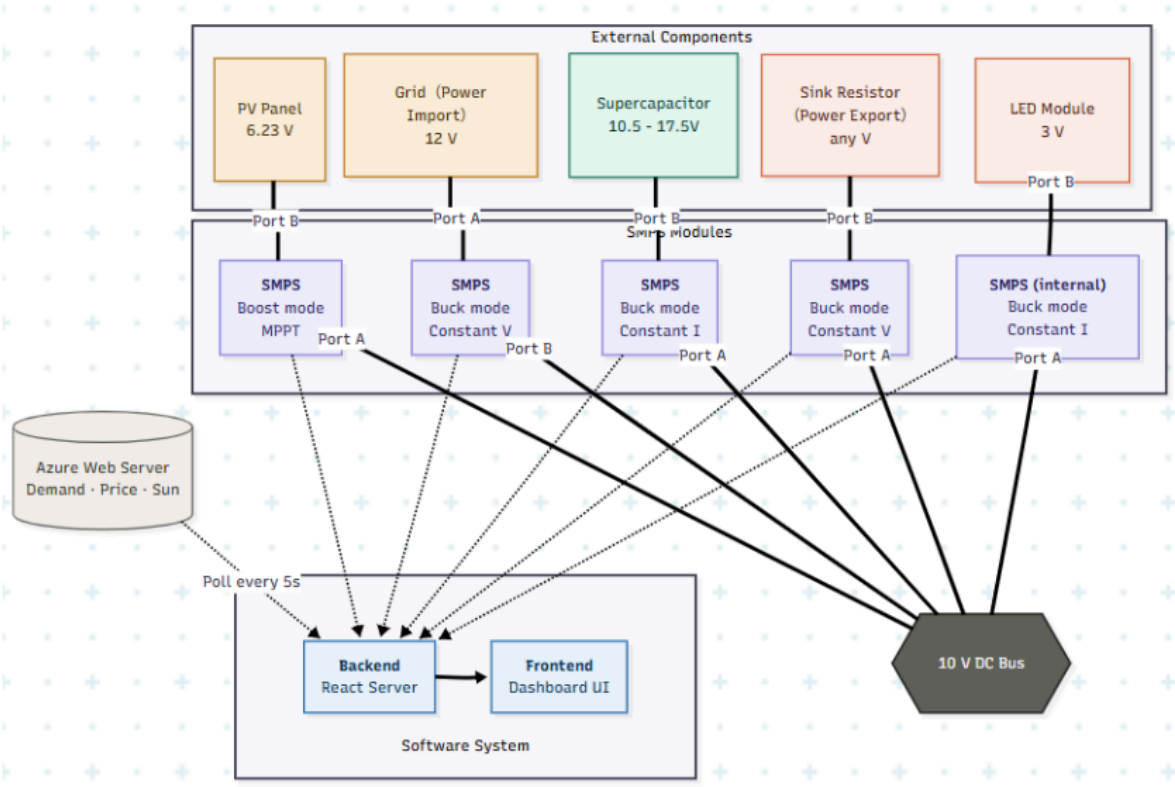


Figure 2.2: Smart Grid system architecture and power flow.

Three properties of this topology recur in later sections. First, every module is autonomous in its bus-regulation role: it acts on its local V_{bus} measurement alone, with the backend supplying only setpoints and economic context. Second, all demo-day power measurements occur at the bus, never at SMPS terminals. Third, the supercapacitor is the only element that stores energy across timescales longer than the bus capacitance, which puts its commanded behaviour at the centre of the economic optimisation.

2.3 Voltage Architecture: 10V Bus, ±0.1V Droop Band and 12V PSU

The voltage architecture is fixed by the operating windows of the PV panel, supercapacitor and grid-interface converters. A 10V nominal bus is the smallest round-number voltage that sits below the supercapacitor charge window, above the PV panel open-circuit voltage of about 8.5V, and reachable from a 12V buck input (justified below). Raising the bus would add switching stress with little benefit; lowering it would reduce the supercapacitor margin and increase the PV boost ratio.

Each droop-controlled module follows a local linear control law of the form

$$I_{\text{module}} = \frac{V_{\text{set}} - V_{\text{bus}}}{R_{\text{droop}}}$$

with sign and saturation set by module role. R_{droop} is sized so each module reaches its rated current at the edge of the ± 0.1V band, bounding the full-load steady-state bus error to $\pm 1\%$ of nominal. This keeps the bus stiff enough for downstream modules while regulation stays local.

The bus voltage choice also affects the scoring of converter losses. Demo-day measurements are taken at the bus, so LED Load and Export losses occur after the measured point and are counted as load or dissipated export power; their efficiency does not affect the score directly. Import losses occur before energy reaches the bus, so every watt lost between the 12V supply and the bus is grid energy missing from the measured balance. Import efficiency therefore gets disproportionate attention in Sections 5 and 11.

The 12V PSU is the lowest practical input that preserves buck regulation while avoiding unnecessary switching loss. For a synchronous buck converter [2],

$$P_{\text{cond}} = D \cdot I_{\text{rms}}^2 \cdot R_{\text{ds,on,HS}} + (1 - D) \cdot I_{\text{rms}}^2 \cdot R_{\text{ds,on,LS}}$$

which reduces to

$$P_{\text{cond}} = I_{\text{rms}}^2 \cdot R_{\text{ds,on}}$$

for matched MOSFETs. The dominant conduction term is therefore independent of both D and V_{in}. Switching loss scales with V_{in} or V_{in}^2:

$$P_{\text{sw}} \approx \frac{1}{2} \cdot V_{\text{in}} \cdot I_{\text{load}} \cdot \left(t_r + t_f\right) \cdot f_{\text{sw}} + \frac{1}{2} \cdot C_{\text{oss}} \cdot V_{\text{in}}^2 \cdot f_{\text{sw}}$$

and inductor ripple shrinks as V_{in} approaches V_{bus}:

$$\Delta I_L = (V_{\text{in}} - V_{\text{bus}}) \cdot D \cdot \frac{T}{L}$$

The binding constraint is therefore dropout, not efficiency: with $D = \frac{V_{\text{bus}}}{V_{\text{in}}}$, duty tends to unity as the input approaches V_{bus}, but a real converter cannot reach $D = 1$ due to minimum off-time, gate-driver refresh and maximum-duty limits. With a maximum duty of about 0.9, regulation is lost below $\frac{V_{\text{bus}}}{0.9} \approx 11$V. A 12V bench supply leaves margin above this floor while staying close enough to the bus to limit switching loss.

Figure 2.3 shows a flat usable region rather than a strong efficiency optimum: across 11 to 15V at a 10V bus and 1.5A load, modelled loss changes by only tens of milliwatts, about 0.2% of delivered power. The sharp collapse below this region is dropout. The 12V point sits on the flat plateau with margin to spare; a future confirmation sweep from 11 to 14V at $V_{\text{bus}} = 10$V would verify the predicted shallow efficiency rise and the dropout cliff.

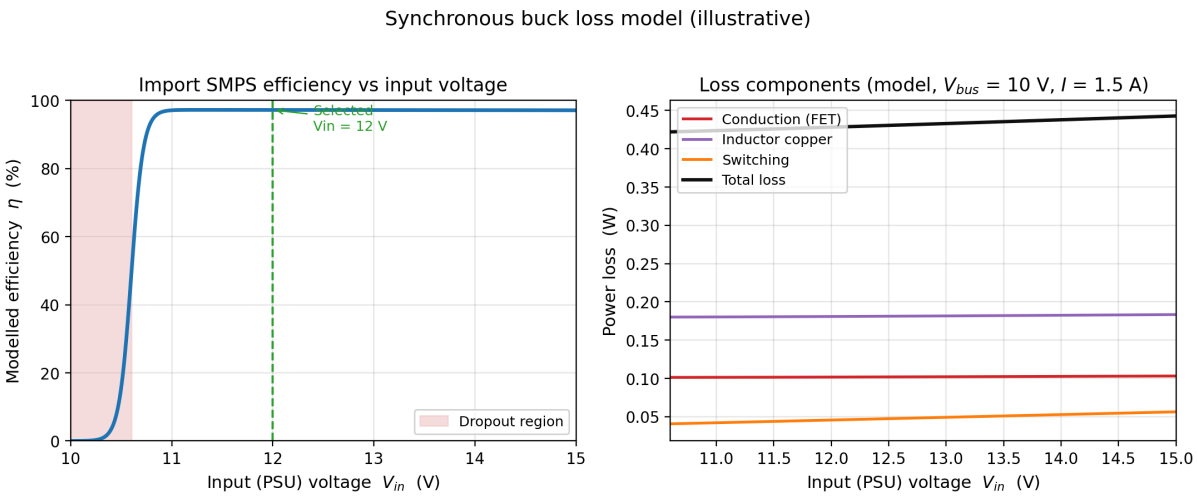


Figure 2.3: Modelled buck efficiency and loss split versus input voltage.

2.4 Coordination Scheme: Symmetric Droop vs Mutex Arbitration

Two architectures were considered. Mutex arbitration assigns control to one module at a time, with a coordinator switching modes (PV-controlled, grid-controlled, supercapacitor-controlled) as conditions change. Symmetric droop, the established primary-control approach for DC microgrids [3], lets every module act at once, each modulating its current from a local V_{bus} measurement against its setpoint.

Mutex arbitration gives tighter steady-state regulation because only one feedback loop is active. It also places the inter-module communication channel inside the safety-critical bus-regulation loop: if comms drop or the coordinator stalls, every module loses its reference and the bus collapses. Mode transitions add voltage transients during handover.

Symmetric droop avoids both failure modes. Each module's loop is local and continuous and needs only its own bus measurement, so a comms failure degrades the economic optimisation but does not threaten bus stability. The droop curves blend, eliminating handover transients, and modules can join or leave the bus without reconfiguring a coordinator. The cost is a small steady-state V_{bus} error, bounded to ± 0.1V by design (Section 2.3).

Symmetry is enforced by mirroring the import and export setpoints about the 10V nominal:

$$V_{\text{set,import}} = 9.9 \text{ V} \qquad V_{\text{set,export}} = 10.1 \text{ V}$$

This creates a dead-band $V_{\text{bus}} \in [9.9, 10.1]$V in which neither grid-facing module acts. Inside the dead-band, the bus is held by the PV module and the supercapacitor — the configuration the algorithm in Section 9 maximises, since no grid interaction means no cost. Outside it, the system falls back to grid import or export with no explicit mode-change logic.

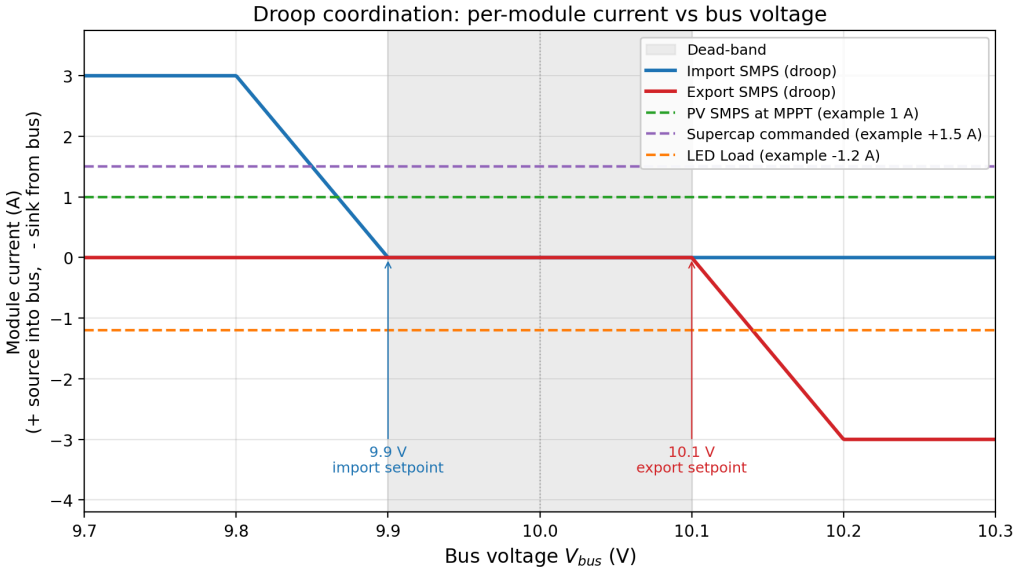


Figure 2.4: Droop curves: module current versus bus voltage.

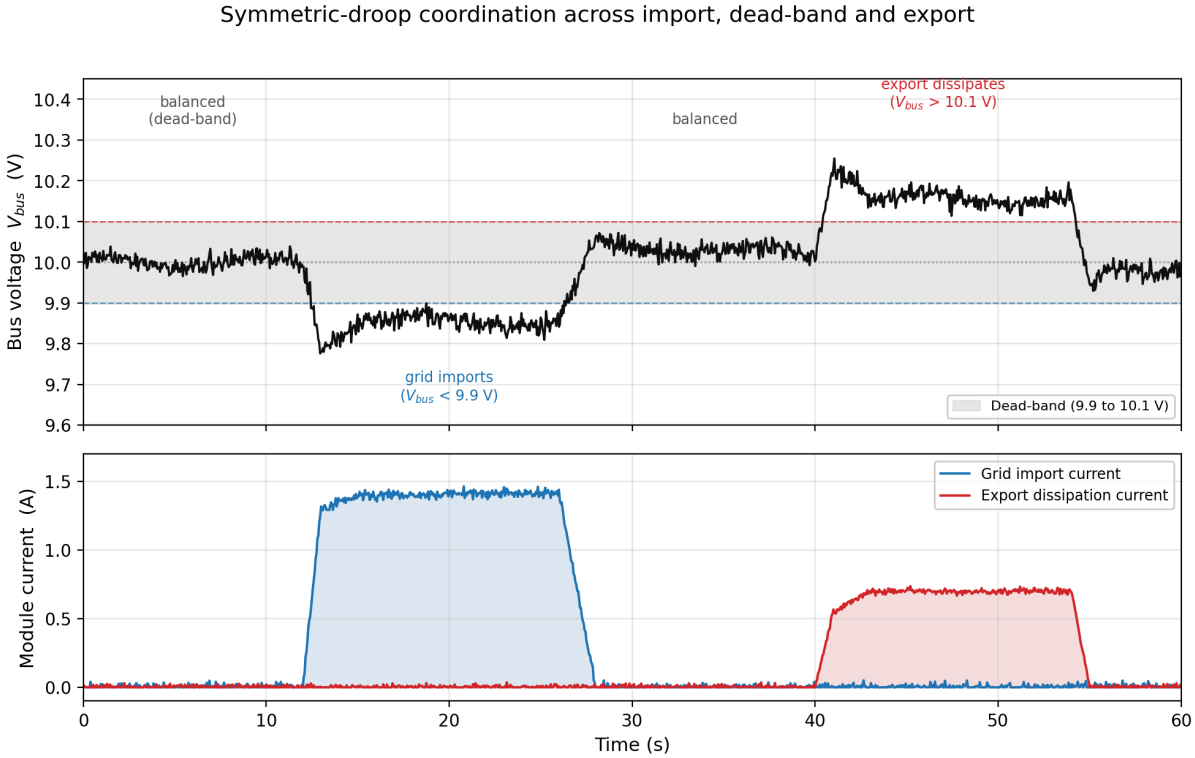


Figure 2.5: Bus voltage and converter currents across import, dead-band and export.

Figure 2.4 and Figure 2.5 show the same behaviour in static and dynamic form: the droop curves define each module's current response, and the load-cycle trace shows the bus moving between import, dead-band and export regimes without central arbitration. Droop is the better architectural choice for the prototype, accepting a small steady-state voltage offset in exchange.

2.5 Communication Protocol

The protocol is provisional and described here at the architectural level only. Each Pico W joins a shared WiFi network (a phone hotspot during bench bring-up, the lab network on demo day) and runs a small HTTP server exposing telemetry and command endpoints in JSON. The

FastAPI backend polls each module at roughly 1 Hz and the Azure-hosted web service every 5 seconds for demand, price and irradiance, computes the dispatch decision (Section 9), and pushes setpoints back over the same HTTP link. The 5-second Azure cadence balances rate-limit considerations against the natural timescale of the demand-and-price signal. The dashboard renders telemetry from the backend and also carries a direct command path to each module (Section 8.3), so operator control survives a backend or database outage.

The protocol sits outside the safety-critical bus-regulation loop. Comms loss leaves every module operating against its last commanded setpoint with its local droop or CC law intact, so the bus keeps regulating. A central safe-default broadcast on the dashboard's Stop action zeroes the commanded current on the supercapacitor and LED Load and restores the grid and export droop targets, so the operator has a one-click recovery without depending on a per-module watchdog. Final framing details (transport, framing, retry policy, error handling) are deferred to module integration testing in Section 11.

3 PV SMPS

3.1 Design & Simulation

3.1.1 Topology Selection

The bench setup uses one PV panel rated at approximately 4W under full bench-emulator illumination at room temperature, with a datasheet open-circuit voltage of 8.5V and short-circuit current of 1.0A. Demo day adds a second identical panel in parallel for roughly double the available harvest, sharing the same MPP voltage but doubling the current. The single-panel characterisation in Section 3.3 transfers directly to the demo-day configuration: the per-panel MPP voltage is unchanged, so the MPPT lookup table, the P&O dwell behaviour and the boost converter's operating duty all stay valid; only the per-panel current at the bus doubles, which sits well under the 1.5A inductor current cap. Under the solar emulator the measured open-circuit voltage was about 7.5V, with the MPP near 5.20V at full irradiance and 5.10V at low irradiance (Figure 3.1). Each curve was taken by stepping the panel voltage from 2V toward the open-circuit knee (with denser sampling around the MPP) at one of six emulator dimmer settings (25% to 100%). Since the MPP voltage (around 5V) is below the 10V bus, energy reaches the bus by stepping up, so the module runs as a boost converter.

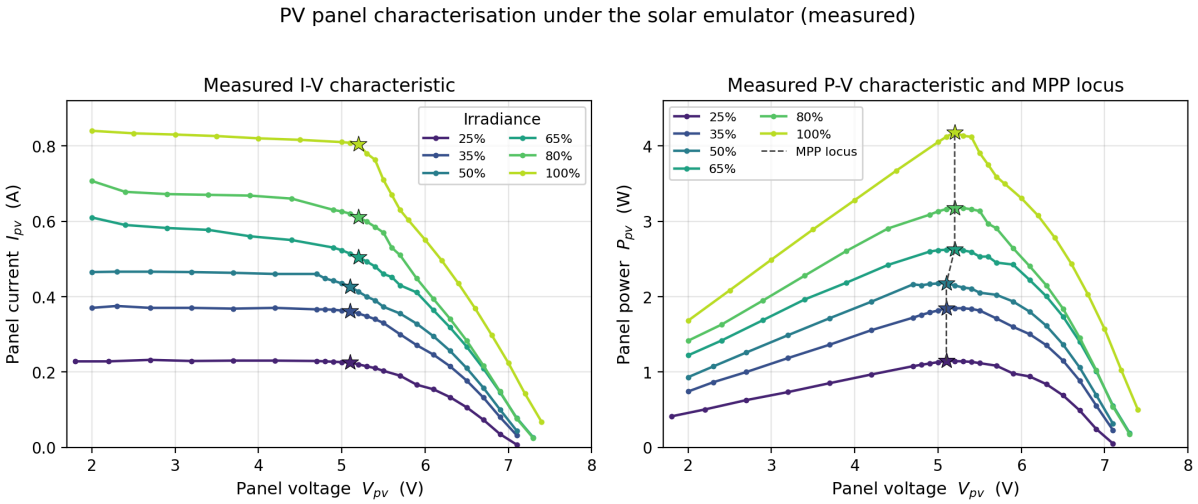


Figure 3.1: Measured PV panel I-V and P-V curves with the MPP locus.

The panel is wired to Port B (lower, variable voltage) and the bus to Port A (fixed 10V), satisfying the hard constraint $V_A > V_B$. The BU/BO switch is set to BOOST, and unlike the buck modules the PWM is not inverted: the firmware writes duty directly rather than the `65536 - pwm_out` form.

Current is sensed by the INA219 across the 0.10 ohm Port B shunt. The shunt is wired in the Port A to Port B sense direction, so the firmware negates the reading; a positive inductor current then corresponds to panel-to-bus boost flow. The controller is cascaded: an outer voltage PI servoes the panel voltage to the commanded V_{mpp}, and an inner current PI drives the duty. The operating point is set by whichever MPPT strategy is active.

3.1.2 MPPT Strategy: Lookup Table vs P&O

Three MPPT approaches were considered: a static constant-voltage operating point, an irradiance-indexed lookup table, and online Perturb and Observe (P&O). All three were prototyped.

The constant-voltage approach is more competitive than expected: Figure 3.2 shows V_{mpp} varying by only 0.10V across the full irradiance range, so a fixed 5.15V setpoint would lose under 1% of available harvest at every characterised point. It is not used alone because the bench sweep is at thermal equilibrium and misses the temperature drift the panel sees under sustained illumination, when V_{mpp} slides with the cell-temperature coefficient. Lookup-table and P&O ship alongside fixed mode; all three are user-selectable from the dashboard so Section 3.3 can compare them under identical conditions.

The lookup table uses the external irradiance signal. The PV module fetches the sun value from the Azure `/sun` endpoint every five seconds, normalises to $[0, 1]$, and interpolates against a piecewise-linear V_{mpp} versus irradiance table built from the sweeps in Figure 3.1. The extracted maximum-power points (Figure 3.2) step from 5.10V at low irradiance (25–50%) up to 5.20V at higher irradiance (65–100%) and are loaded directly into the firmware table. The 0.10V step direction matches the textbook silicon-cell ascent of V_{mpp} with irradiance, though the magnitude is small enough that the lookup is nearly flat. Settling is immediate because the new V_{mpp} is a step input to the existing outer voltage PI. The drawbacks are the upfront characterisation effort and dependence on the web link; if the link drops, the firmware holds the last good irradiance value instead of guessing.

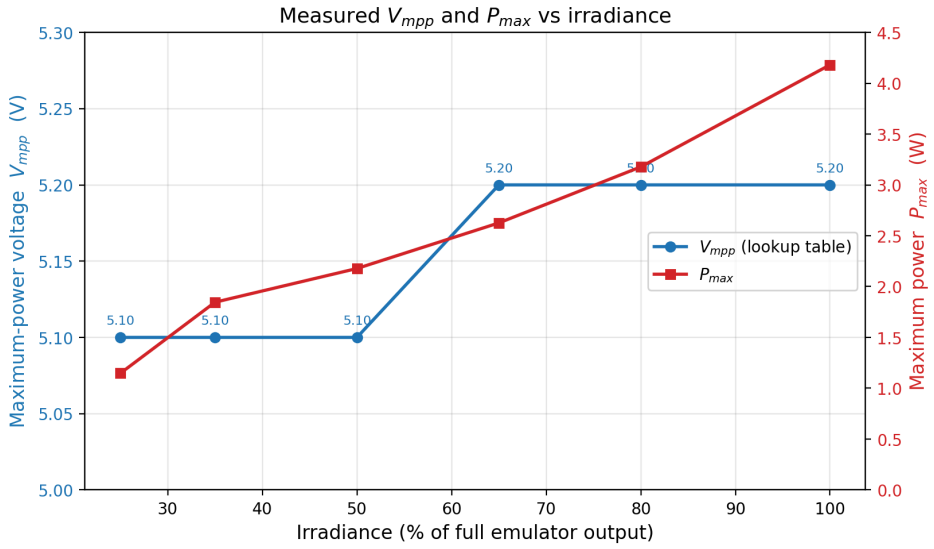


Figure 3.2: Measured Vmpp and Pmax versus irradiance: the MPPT lookup table.

P&O makes no assumption about V_{mpp}. It perturbs the commanded V_{mpp} by a small fixed step, measures the resulting power averaged over a settling window, compares it to the previous window, and either keeps or reverses direction. With no external irradiance signal needed, P&O is more tolerant of web-link outages or unmodelled panel behaviour, at the cost of constant oscillation around the peak and slower convergence after rapid irradiance steps. The P&O literature [4] warns that one-sample-per-perturbation implementations collapse into noise-driven random walks when per-sample noise exceeds the curvature of $P(V_{\text{mpp}})$ near the peak. The bench noise floor in Section 3.2 made this a real problem.

Shipping all three gives a clean comparison under identical conditions and a fallback path if the web link fails: LUT drops to Fixed mode, and P&O stays available as an autonomous fallback.

3.2 Testing & Implementation

Three concerns dominated bringing up the PV SMPS: noise on the measured panel power that masked operating-point differences, P&O behaviour against that noise, and a clean framework for comparing the MPPT modes. Each is described in turn.

3.2.1 Bench Noise Treatment

Raw $P_{\text{panel}} = V_{\text{panel}} \cdot I_{\text{L}}$ telemetry showed several-hundred-milliwatt peak-to-peak jitter at the 1 kHz tick, swamping the mode-to-mode differences Section 3.3 needed to resolve. The three-layer attenuation derived in Section 7 (INA219 hardware averaging, four-fold Pico ADC oversample, and a software EMA at $\alpha = 0.05$) was applied. The filtered values feed only the dashboard and the P&O comparison; the inner current PI and safety guards still use raw single-sample readings, leaving loop tuning unchanged and keeping trip response instantaneous.

3.2.2 P&O Implementation

P&O uses a two-timescale structure. The outer voltage PI tracks the commanded V_{mpp} at 100 Hz; perturbations occur on a 300 ms cadence (T_{dwell}), roughly five times the outer-loop settling time, so the power measured at the end of a dwell reflects steady state, not the PI transient. The comparison power is the filtered $V_{\text{panel,lp}} \cdot I_{\text{L,lp}}$ accumulated over the second half of each dwell and averaged. Two-stage filtering (EMA then dwell average) brings the per-comparison noise floor under 10 mW, well below the $P\big(V_{\text{mpp}}\big)$ curvature near the peak.

A direction-flip rule with a dead-band stops the three-point oscillation degenerating into a noise-driven random walk. If dwell-averaged power dropped by more than $\varepsilon = 20$ mW versus the previous dwell, direction reverses; otherwise it continues. The commanded V_{mpp} is hard-clamped to 4.75V to 5.5V, a narrow window centred on the measured 5.10 to 5.20V MPP locus (Figure 3.2), so an algorithm fault cannot walk the operating point into open-circuit or short-circuit territory. On entry to P&O mode the state is reseeded from the last commanded fixed-mode V_{mpp}, avoiding a stale value.

3.2.3 Mode Comparison Support

A three-second block average of P_{panel}, restarting on each mode switch, was added to the firmware telemetry so the operator can read steady-state delivered power for each MPPT mode straight from the dashboard; the live trace captures tracking noise separately. The same block-average pattern is reused in Sections 4 and 5.

3.3 Evaluation & Optimisation

Two evaluation axes are reported: MPPT mode comparison, and SMPS conversion efficiency.

3.3.1 MPPT Mode Comparison

Each MPPT mode is tested across the lookup table's irradiance range, with the lab dimmer set at 25%, 50%, 75% and 100% fractional sun. At each setting the three-second block-averaged P_{panel} is recorded in fixed mode (commanded at the table's V_{mpp}), LUT mode (tracking the live sun signal), and P&O mode (started from the LUT seed). $V_{\text{mpp,target}}$ and v_{panel} are logged throughout so the mode-switch transient can be inspected. The primary metric is steady-state delivered power; the secondary metric is the mode-change transient.

Mean recorded powers per mode and irradiance are summarised in Figure 3.3 and tabulated below.

Irradiance	Fixed (W)	LUT (W)	P&O (W)	Spread (mW)
25%	0.767	0.770	0.770	3
50%	1.933	1.932	1.933	1
75%	2.845	2.841	2.841	4
100%	3.577	3.562	3.550	27

Table 3.1: Recording-phase mean panel power per MPPT mode and irradiance.

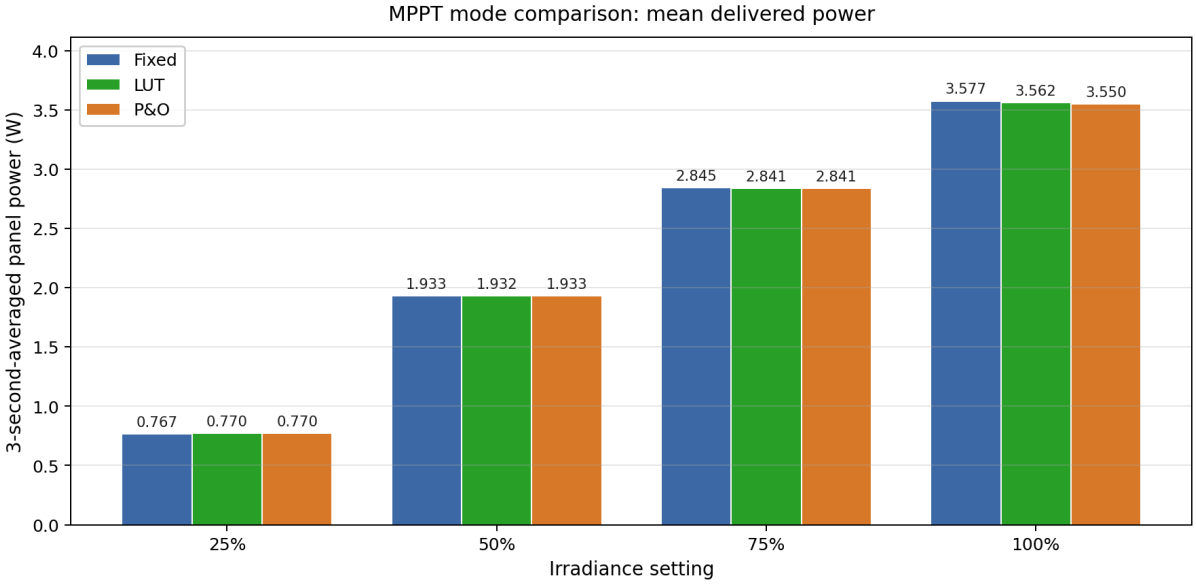


Figure 3.3: Mean delivered panel power by MPPT mode at four irradiance settings.

The three modes converge to within a few mW at every irradiance; the worst-case spread of 27 mW at 100% is under 1% of the 3.55 W operating point. This is what the construction predicts: P&O hunts to the same peak the LUT is built from, and Fixed sits within 0.10V of V_{mpp} across the full range (Figure 3.2), so all three settle close to the same operating point. At thermal equilibrium under the bench emulator there is no steady-state harvest advantage of P&O over LUT or Fixed.

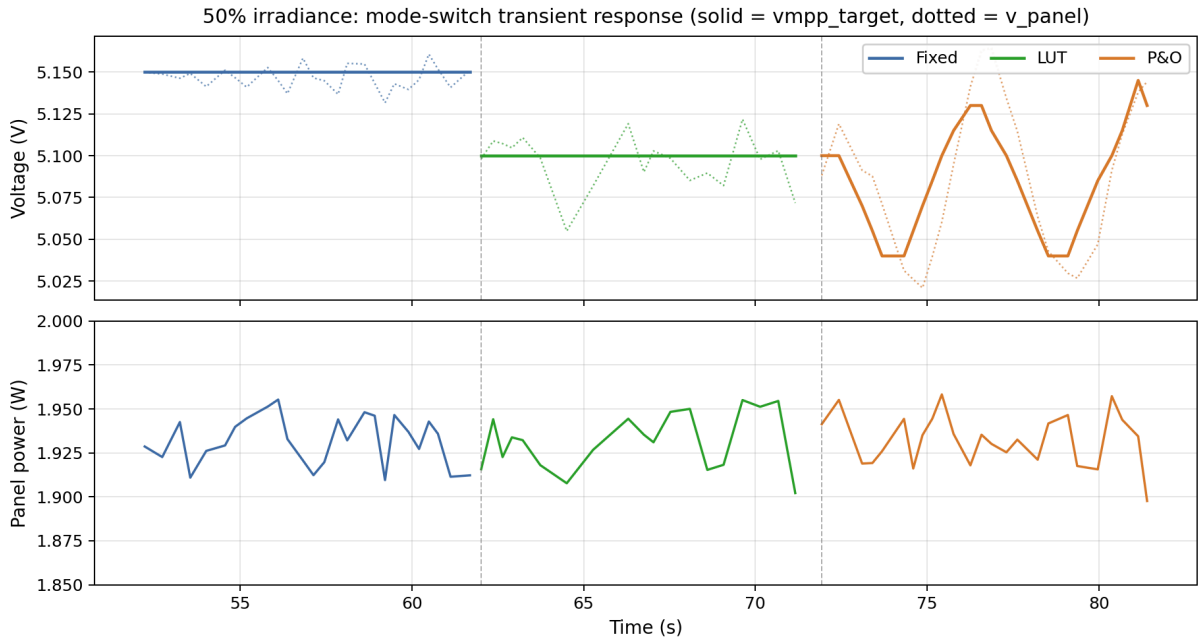


Figure 3.4: Mode-switch transient at 50% irradiance: commanded V_{mpp} (solid), measured v_{panel} (dotted), and resulting P_{panel}.

Figure 3.4 shows the mode-switch transient at 50% irradiance. Fixed and LUT hold their commanded V_{mpp} flat at 5.15V and 5.10V, with v_{panel} tracking inside the noise envelope. P&O shows the expected three-point hunting oscillation around the peak, walking V_{mpp} up and down in a bounded cycle; the resulting panel power stays inside the same envelope as the other two modes, confirming the algorithm locks onto the peak rather than drifting.

The bench shows no steady-state advantage for P&O, but it does confirm the algorithm tracks correctly and that the oscillation stays inside the noise envelope without drift. The temperature effect the bench cannot reproduce is what matters for sustained operation: as the panel warms under continuous illumination the silicon-cell temperature coefficient slides V_{mpp} down by several mV per °C. Fixed and LUT are open-loop with respect to temperature and accumulate a steady-state error that grows with run time. P&O is closed-loop on power and walks the operating point to the present V_{mpp}, so it has zero temperature-driven offset. The shipped default is therefore P&O, with Fixed and LUT retained as user-selectable fallbacks so a P&O fault degrades to a working but lower-tracking mode instead of collapsing harvest.

3.3.2 Measured SMPS Efficiency

PV conversion efficiency is defined as $\eta_{\text{PV}} = \frac{P_{\text{bus,out}}}{P_{\text{panel,in}}}$, with $P_{\text{panel,in}} = V_{\text{panel}} \cdot I_{\text{L}}$ from firmware (both factors are Port B) and $P_{\text{bus,out}} = V_{\text{bus}} \cdot I_{\text{bus,out}}$ from a DMM in series with the PV-to-bus wire (Port A is uninstrumented). Figure 3.5 and the table below show the 48-point sweep: four irradiance settings (25%, 50%, 75%, 100%) at twelve bus-voltage points each, with the panel pinned at its MPP ($V_{\text{panel}} \approx 5.15$ V, from Figure 3.2). The boost duty $D = 1 - \frac{V_{\text{panel}}}{V_{\text{bus}}}$ spans 0.28 to 0.59 across the sweep, staying inside the flat boost plateau.

V_{bus} (V)	Duty D	η_{PV} at 25%	η_{PV} at 50%	η_{PV} at 75%	η_{PV} at 100%
7.11	0.28	96.56%	96.69%	97.17%	97.68%
7.61	0.32	96.36%	96.79%	96.82%	97.89%
8.11	0.37	96.62%	96.57%	96.98%	97.74%
8.61	0.40	96.52%	96.44%	96.73%	98.00%
9.12	0.44	96.16%	96.63%	96.79%	97.86%
9.61	0.46	96.69%	96.34%	97.11%	98.10%
10.21	0.50	96.56%	96.80%	97.15%	98.02%
10.63	0.52	96.57%	96.68%	97.28%	97.79%
11.07	0.54	96.18%	96.75%	96.90%	98.10%
11.63	0.56	96.37%	96.51%	96.92%	98.08%
12.06	0.57	96.32%	96.88%	96.98%	97.83%
12.55	0.59	96.66%	96.84%	96.81%	97.77%

Table 3.2: Measured PV SMPS efficiency across bus voltage and irradiance.

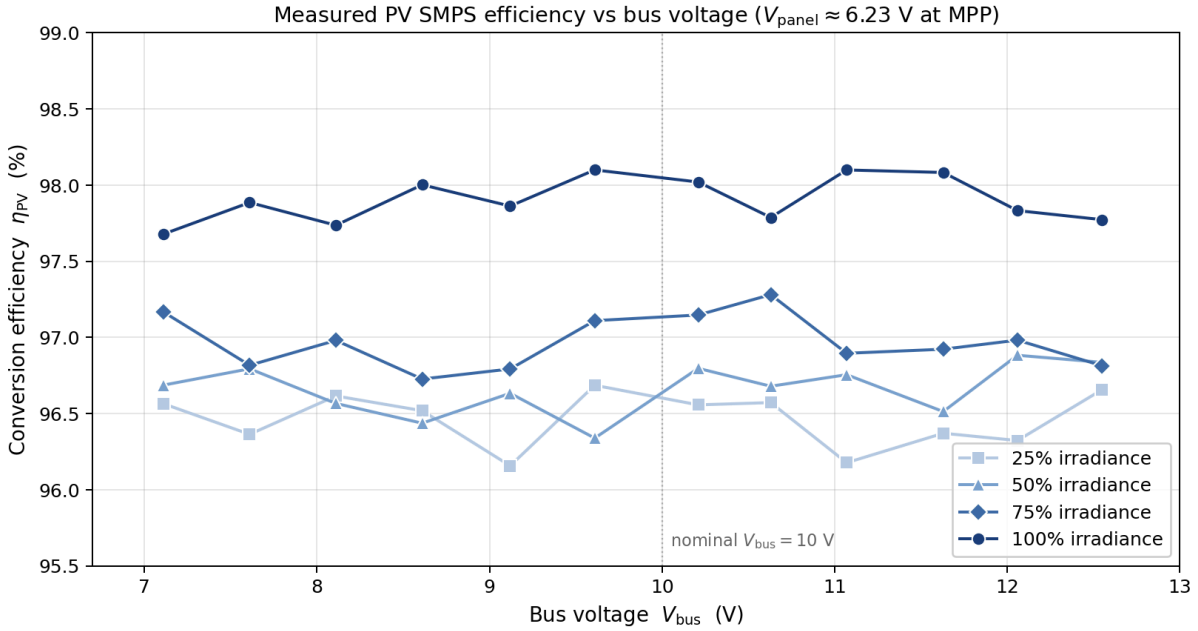


Figure 3.5: Measured PV SMPS efficiency versus bus voltage at four irradiances.

Two features stand out. First, each η_{PV} trace is flat with bus voltage. The four curves vary by no more than $\pm 0.3\%$ across their twelve points, with no monotonic trend against V_{bus}, matching the flat plateau of the boost η-versus-D characteristic over the $D = 0.28$ to 0.59 range. Fig. 5 of [5], reproduced in Figure 3.6, shows this plateau across moderate duty cycles and a sharp CCM cliff beyond $D \approx 0.7$. At the supercapacitor ceiling $V_{\text{bus}} = 17.5$V the reachable $D_{\max} = 1 - \frac{5.15}{17.5} \approx 0.71$ sits at the cliff threshold, so the supercapacitor-full corner is the one regime where degraded efficiency would be expected; elsewhere the plateau represents converter behaviour across its operating window.

Second, the per-irradiance means rise monotonically: 96.5% at 25%, 96.7% at 50%, 97.0% at 75% and 97.9% at 100%. The 1.4 percentage-point spread matches the boost loss model. Fixed

gate-drive and switching overhead are independent of throughput, so they consume a larger fraction at low panel power: 100% irradiance gives about 2.3W out, 25% gives about 0.8W, and the same fixed overhead is a bigger share of the lower throughput. The irradiance-to-irradiance gap sits well above the per-point scatter, so the monotonic ordering is statistically clean rather than a noise artefact.

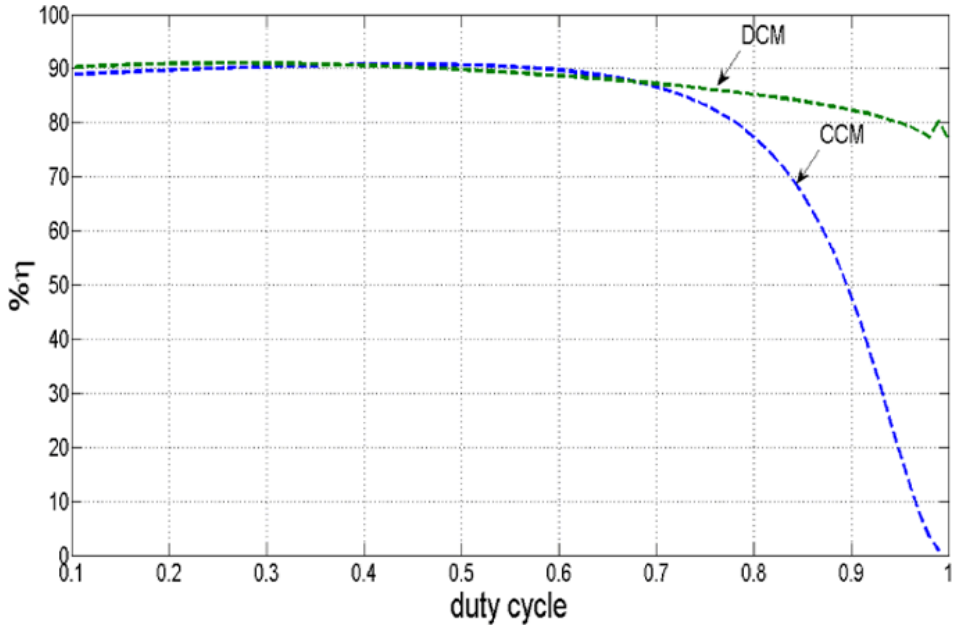


Figure 3.6: Boost efficiency versus duty cycle in DCM and CCM, from Fig. 5 of [5].

PV conversion efficiency stays within 96 to 98% across all four irradiance levels, rising 1.4 percentage points from 25% to 100% irradiance with no meaningful dependence on bus voltage. Import efficiency does enter the score under the dual-meter framework of Section 2.3 and rises monotonically with V_{bus} (Section 5.3.2), so the import lens marginally favours a higher bus voltage; the PV lens is neutral. The $V_{\text{bus}} = 10$V choice rests on hardware constraints (supercapacitor lower bound, PV panel open-circuit voltage, buck dropout), with both bus-relevant efficiency curves consistent with that pick.

4 Supercapacitor SMPS

4.1 Design & Simulation

4.1.1 Boost Topology and Boot-with-Empty-Capacitor Rationale

The supercapacitor is the only variable-voltage element on the storage side, ranging from 10.5V when depleted to 17.5V when full. The same Port A above Port B transfer rule that fixed the PV topology dictates the port assignment here (the shared power stage in Figure 2.1): the capacitor is the higher and variable rail and must sit on Port A, with the fixed 10V bus on Port B. Energy flows from bus to capacitor (B to A) when charging and from capacitor to bus (A to B) when discharging, so the module is a bidirectional SMPS.

The SMPS is set to BOOST mode, which is the load-bearing decision for robustness. In BOOST mode, the on-board circuitry draws from Port B, the bus, which is always energised by the grid and PV modules. The SMPS therefore boots and stays alive even when the capacitor is fully empty at 0V. The alternative BUCK setting would power the bus from Port A, the capacitor, and would trip the SMPS whenever the capacitor fell below the 6V to 7V minimum support voltage. The current sign convention follows the PV module: a positive inductor current means charging, a negative current means discharging, and the dashboard command i_{cmd} is signed accordingly.

4.1.2 Operating Window 10.5 to 17.5 V

The usable voltage window is determined by two hard limits, illustrated in the stored-energy curve in Figure 4.1. The ceiling $V_{\text{cap,max}} = 17.5\text{V}$ is the SMPS port limit and sits under the 18V capacitor rating. The floor $V_{\text{cap,min}} = 10.5\text{V}$ is a limit point; below that the capacitor cannot satisfy $V_{\text{A}} > V_{\text{B}}$ against the 10V bus. Within this window the capacitor stores energy as $E = \frac{1}{2}CV^2$ with $C = 0.5\text{F}$ (two 0.25F capacitors in parallel).

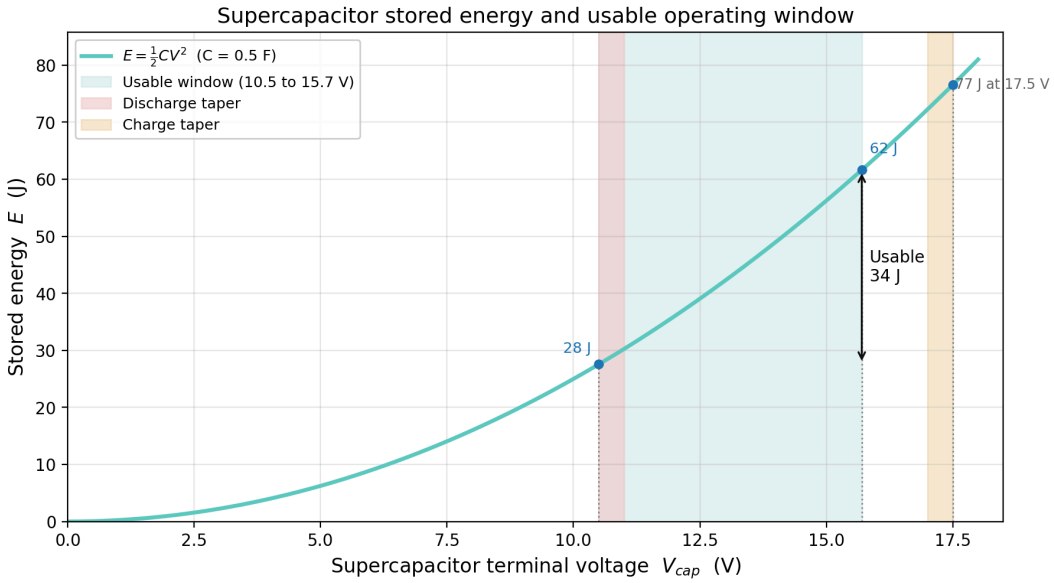


Figure 4.1: Supercapacitor stored energy and usable operating window.

In firmware the practical voltage ceiling is lowered to 15.7V, to reduce the impact of noise overshooting the voltage limit if 17.5V was used instead. The current is reset to 0 whenever the capacitor approaches within 0.5V of the upper and lower limits. The energy genuinely

transferred between 10.5V and 15.7V is approximately 34J of the 49J the full window could hold. A significant complication is the capacitor's approximately 4 ohm equivalent series resistance (ESR): immediately after a current step the voltage ripple is dominated by the resistive drop and charge redistribution, so an accurate voltage can only be determined after some settling time. This directly shapes the cycle-test settle phases described in Section 4.3, which read the open-circuit voltage only after a deliberate pause. Peak current is clamped to 0.60A, leaving approximately 21% headroom under the 0.76A SMPS hardware limit [6].

4.1.3 Command-Driven CC: Rationale for the Departure

Unlike the grid and export modules, the supercapacitor is not involved in droop. The bus voltage is already regulated by the grid and export droop pair (Section 2.4). Instead the capacitor is a commanded current source and sink: it tracks a signed current given by input from the economic algorithm. This is the correct implementation for storage where it allows arbitrage: charging when energy is cheap and discharging when it is expensive, rather than reacting passively to instantaneous bus voltage.

The control law is an inner current PI only. The current limits implemented are a maximum or minimum at ± 0.60 A, a linear taper to 0 near the voltage limits, and a bus-health taper that reduces charging demand when the bus voltage sags so the capacitor does not pull too much grid current. The detailed feedforward and anti-runaway treatment of this inner loop is described in Section 4.2.

4.2 Testing & Implementation

Two implementation matters specific to the supercapacitor module are reported here: the same layered noise treatment described for the PV module in Section 3.2, and the inner-loop feedforward that keeps the synchronous boost stable.

4.2.1 Inner-Loop Feedforward and Bus-Droop Backoff

The inner current PI carries a boost feedforward term equal to the equilibrium duty cycle, $D_{\text{ff}} = 1 - \frac{V_{\text{bus}}}{V_{\text{cap}}}$. At this duty the inductor sees zero net volt-seconds per switching cycle, so its current holds constant and the integrator only corrects the residual. Without the feedforward the integrator must synthesise the entire steady-state duty, and when the command steps to zero the PI can transiently drive the duty all the way to its floor. In this synchronous boost a zero duty clamps the high-side MOSFET on, dumping capacitor energy back into the bus and crashing the rail.

The feedforward uses the nominal 10V bus value, not the measured V_{bus}. Feeding the measurement back in creates a positive-feedback loop with gain near 1.7: a bus sag raises the feedforward duty, which draws more current from the bus, sagging it further. Using the constant nominal value breaks the loop, and the integrator absorbs any real offset between nominal and actual bus voltage as a small steady-state correction. A complementary bus-health taper scales the charge command down as the bus falls from 9.5V toward 8.5V, so the capacitor stops competing with the grid module for current during a shared sag; discharging is left untapered because pushing current into a sagging bus helps. A hard charge cutoff at 15.7V is the final backstop above the soft taper.

4.3 Evaluation & Optimisation

The supercapacitor evaluation answers two questions: what is the round-trip energy efficiency η_{RT}, and how does it split into η_{charge} and $\eta_{\text{discharge}}$? Both are answered from a single charge-and-

discharge cycle, using the capacitor's known capacitance as an internal calibration reference. The same dataset supplies the efficiency-versus-current curve needed by Section 11.

4.3.1 Energy Balance Method

For a complete cycle from V_{low} up to V_{high} and back, three energies are measured. Bus-side input over the charge phase is $E_{\text{in}} = \int V_{\text{bus}} \cdot I_{\text{bus}} dt$. Bus-side output over the discharge phase is $E_{\text{out}} = \int V_{\text{bus}} \cdot |I_{\text{bus}}| \; dt$. Capacitor-side stored energy at three reference points is $E_{\text{cap}} = \frac{1}{2} C V_{\text{cap}}^2$ from the open-circuit capacitor voltage, captured before charging ($E_{\text{cap,start}}$), between charge and discharge ($E_{\text{cap,peak}}$) and after discharging ($E_{\text{cap,end}}$). The three efficiencies follow:

$$\eta_{\text{charge}} = \frac{E_{\text{cap,peak}} - E_{\text{cap,start}}}{E_{\text{in}}}$$

$$\eta_{\text{discharge}} = \frac{E_{\text{out}}}{E_{\text{cap,after hold}} - E_{\text{cap,end}}}$$

$$\eta_{\text{RT}} = \frac{E_{\text{out}}}{E_{\text{in}}}$$

The discharge denominator uses $E_{\text{cap,after hold}}$ rather than $E_{\text{cap,peak}}$ so the efficiency reflects only the round-trip path through the converter, with the optional zero-current hold isolated as a separate self-discharge term $\eta_{\text{self}} = \frac{E_{\text{cap,after hold}}}{E_{\text{cap,peak}}}$ (Section 4.3.4). With a zero-length hold the two are identical and $\eta_{\text{RT}} = \eta_{\text{charge}} \cdot \eta_{\text{discharge}}$. The round-trip metric η_{RT} is itself bus-side only and immune to the capacitor-side measurement issues discussed below.

No current sensor is required on the capacitor side: E_{cap} comes from one voltage measurement and the known capacitance $C = 0.5\text{F}$ (two 0.25F capacitors in parallel). This avoids the cascading-efficiency problem of multi-stage characterisation and keeps the test self-contained inside the supercapacitor firmware.

4.3.2 Cycle Test State Machine

The test is a state machine inside the supercapacitor firmware, triggered from one dashboard button (Figure 4.2). Phases run in sequence: precharge (drive V_{cap} down to V_{low} if above), settle 1 (zero current for 500 ms, capture $E_{\text{cap,start}}$), charge (constant current I_{charge} up to V_{high}), settle 2 (capture $E_{\text{cap,peak}}$), hold (zero-current dwell for the operator-specified duration, capture $E_{\text{cap,after hold}}$ at the end), discharge (constant current $I_{\text{discharge}}$ back to V_{low}), settle 3 (capture $E_{\text{cap,end}}$), done (compute and expose the four efficiencies). A hold duration of zero falls through the hold phase on its first tick, so the cycle shape and efficiency numbers are identical to the no-hold case used in Section 4.3.3. Each settle phase averages V_{cap} over only its last 200 ms, skipping the early ESR relaxation transient. Integration of E_{in} and E_{out} runs at the full 1 kHz tick rate so the displayed efficiencies are not limited by any external sample rate.

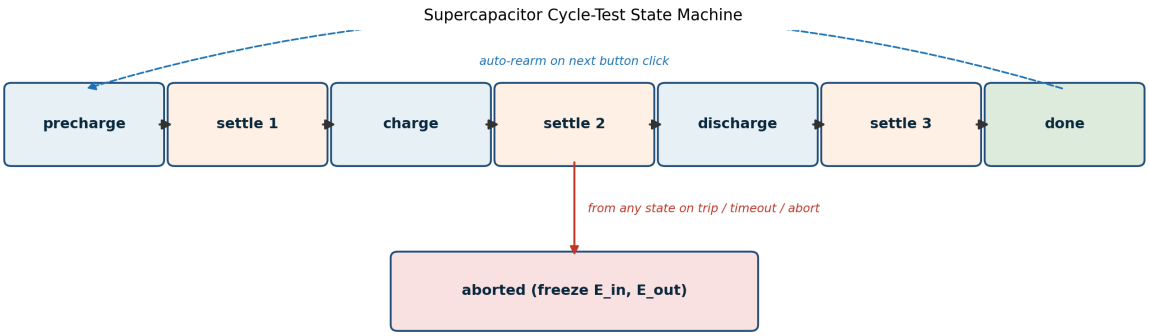


Figure 4.2: Supercapacitor cycle-test state machine.

Three state-machine design decisions are worth noting. First, the endpoint is $V_{\text{low}} = 10.5\text{V}$ instead of 11V, letting the discharge current taper to zero below 11V. The energy integral captures the tapered region, and $\eta_{\text{discharge}}$ reflects performance averaged across the operational window; an 11V endpoint would understate the discharge contribution to round-trip loss. Second, the test bypasses the user-i_{cmd} path and writes 0 into the command dictionary at start, so the capacitor goes idle when the test ends. Third, auto-rearm: after the done phase the firmware returns to normal operation, so the next test starts with one button click without an intermediate reset.

4.3.3 Measured Results

The cycle test was run as two sweeps, each three times: a current sweep at 0.1, 0.2, 0.3, 0.4, 0.5 and 0.6A with the bus at 10V, and a bus-voltage sweep at 8.0, 8.5, 9.0, 9.5 and 10.0V at 0.2A. Figure 4.3 shows a representative cycle at 10V, 0.2A. V_{cap} ramps from 10.5V to the 15.7V cutoff over a 19-second charge and back over a 17-second discharge; the bus-side integrals reach $E_{\text{in}} = 38.2$ J on charge and $E_{\text{out}} = 33.7$ J on discharge. The capacitor-side energy swing $\frac{1}{2}C\left(V_{\text{high}}^2 - V_{\text{low}}^2\right) \approx 34$ J matches the design estimate.

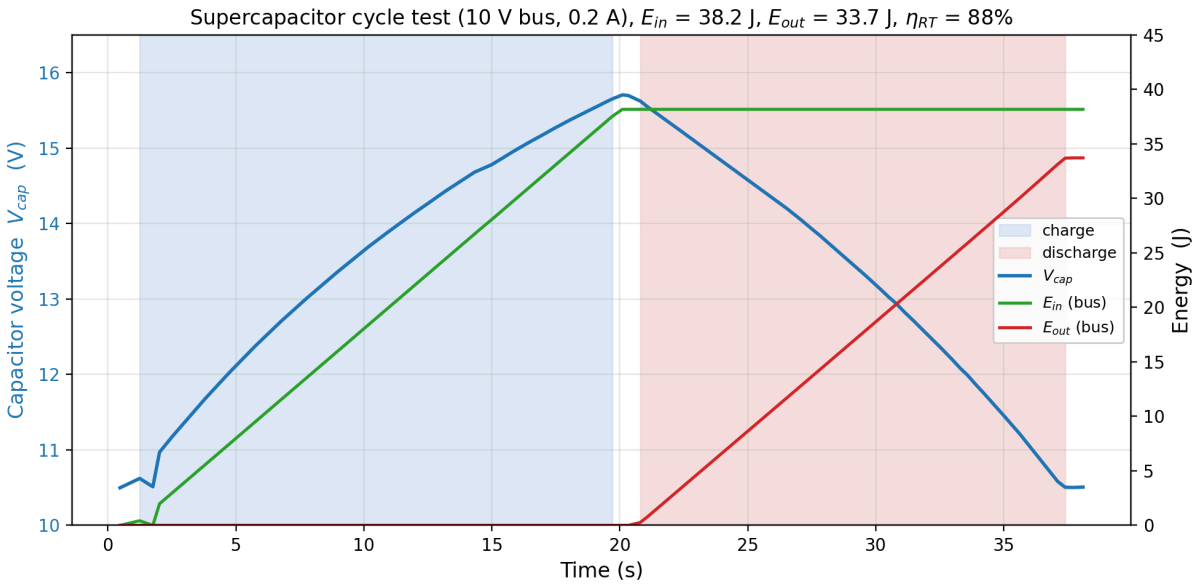


Figure 4.3: Measured supercapacitor charge and discharge cycle at 10V, 0.2A.

Round-trip efficiency is $\eta_{\text{RT}} = \frac{E_{\text{out}}}{E_{\text{in}}}$, with both energies measured at the bus, so the figure is independent of any capacitor-side assumption. Figure 4.4 plots it against current and bus voltage with all three runs per point. The headline is approximately 88% round-trip, slightly above

the conservative 70 to 85% pre-test estimate and stable across the operating range. Current dependence is weak and slightly negative across 0.1 to 0.6A (89.7% at 0.1A falling to 86.9% at 0.5A, with 87.4% at 0.6A within run-to-run scatter), matching conduction loss increasing with current. Bus-voltage dependence is weaker and largely within scatter, with a mild rise toward higher bus voltage (87.3% at 8V to 88.3% at 10V) consistent with the smaller boost ratio.

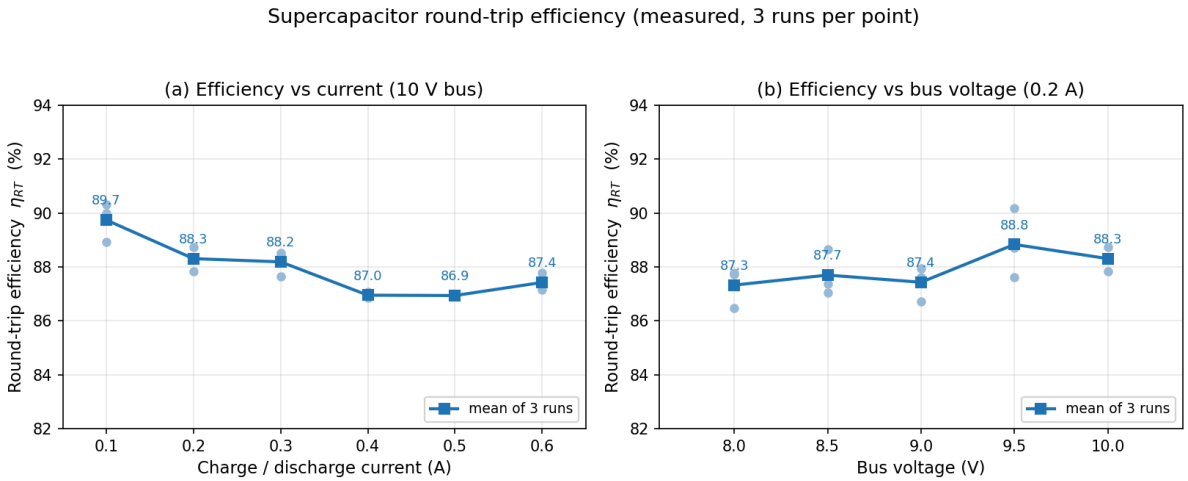


Figure 4.4: Measured round-trip efficiency versus current and versus bus voltage.

Charge current	Mean η_{RT}, 3 runs
0.1 A	89.7%
0.2 A	88.3%
0.3 A	88.2%
0.4 A	87.0%
0.5 A	86.9%
0.6 A	87.4%

Table 4.1: Supercapacitor round-trip efficiency by charge current at 10 V bus.

Bus voltage	Mean η_{RT}, 3 runs
8.0 V	87.3%
8.5 V	87.7%
9.0 V	87.4%
9.5 V	88.8%
10.0 V	88.3%

Table 4.2: Supercapacitor round-trip efficiency by bus voltage at 0.2 A.

The per-phase split into η_{charge} and $\eta_{\mathrm{discharge}}$ is not reported because the data is unreliable on this capacitor. The open-circuit energy $\frac{1}{2}CV^2$ needs to be measured at each settle phase, but ESR-driven charge redistribution stops the voltage holding constant. The discharge efficiency routinely came out above 100%, which is impossible. $\eta_{\mathrm{RT}} = \frac{E_{\mathrm{out}}}{E_{\mathrm{in}}}$ is computed from bus-side measurements only and is immune; so is the figure carried into Section 11. The capacitor-side voltage is good enough for the state-of-charge display but not for a charge-versus-discharge split.

4.3.4 Self-Discharge and Hold-Duration Dependence

The cycle test was extended with a zero-current hold phase between charge and discharge, and a sweep at 60, 120, 180, 240 and 300 seconds (three runs each, 0.2A at 10V bus) was run to characterise self-discharge. Figure 4.5 shows the drop in round-trip efficiency, and Figure 4.6 shows the underlying open-circuit relaxation.

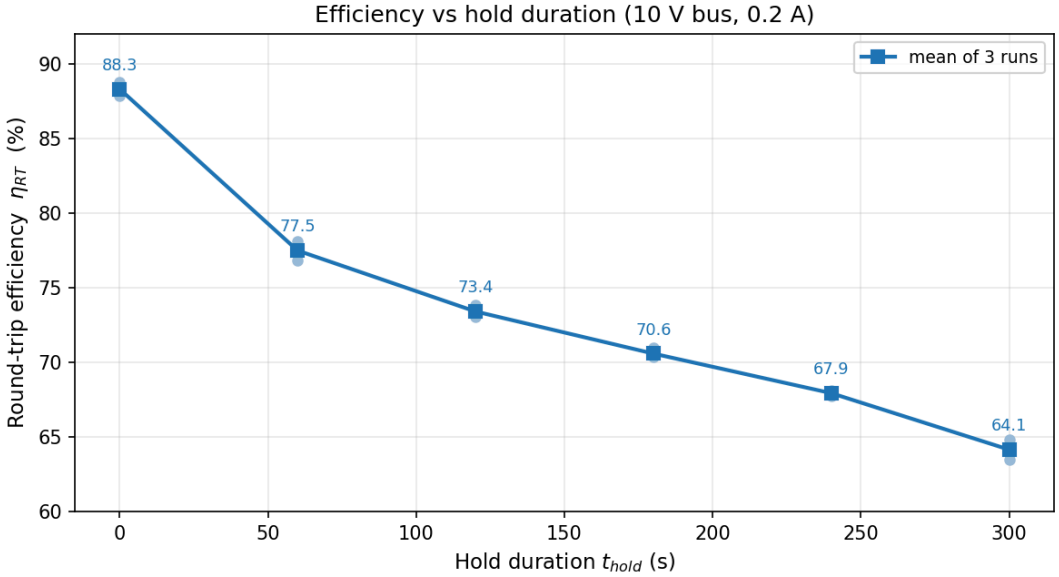


Figure 4.5: Round-trip efficiency versus zero-current hold duration.

η_{RT} falls from 88.3% with no hold to 64.1% after 300 seconds at peak charge. The biggest step is the first minute (88.3% to 77.5%); subsequent minutes are linear at roughly 3% per minute. Since $\eta_{\mathrm{RT}} = \frac{E_{\mathrm{out}}}{E_{\mathrm{in}}}$ is measured at the bus, this is real energy lost, not a measurement artefact.

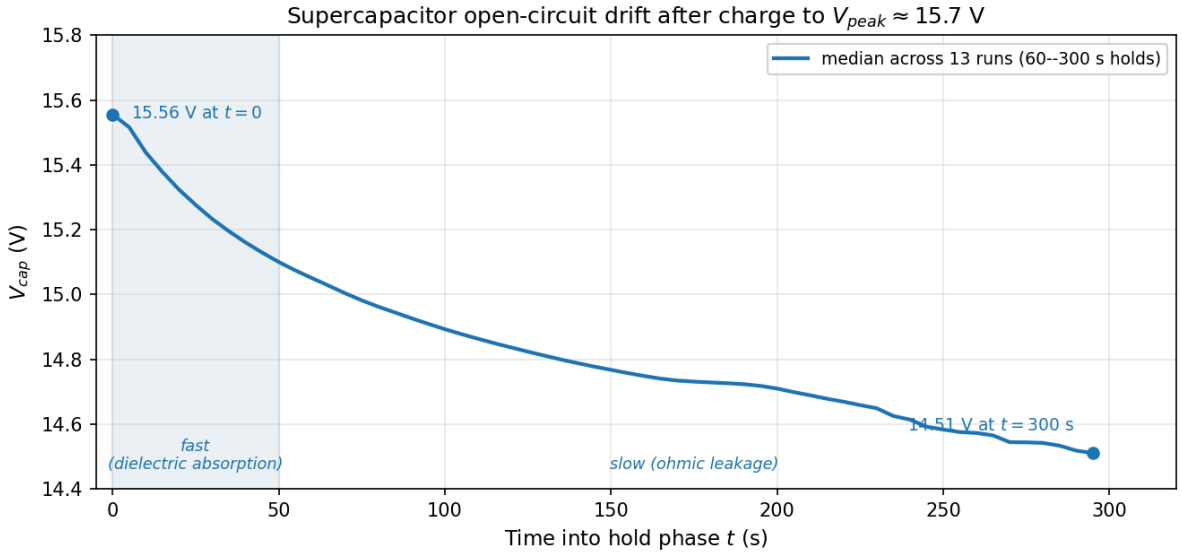


Figure 4.6: Open-circuit V_{cap} drift after charge: master curve across 13 hold runs.

Figure 4.6 explains the efficiency drop. The median V_{cap} across all 0.2A hold runs collapses onto a single relaxation curve, so one mechanism dominates: a fast component of about 0.5V over the first 50 seconds, from dielectric absorption (charge redistribution into deeper dielectric layers [7], the reason Section 4.1 inserts a settle window before each OCV capture), and a slow

component of 2 to 3 mV per second afterwards from ohmic leakage through the dielectric. All hold lengths give the same curve, so the relaxation can be fit once and reused for any hold time.

The implication for the economic dispatch algorithm in Section 11 is direct: the supercapacitor should be charged just-in-time. Even a few minutes of hold time at peak state of charge severely decreases the round-trip efficiency. It is better to delay charging by a few minutes than to charge ahead and let leakage decrease the voltage.

5 Import & Export SMPS

5.1 Design & Simulation

5.1.1 Topology Choice

Both grid-facing modules are configured as synchronous buck converters with the BU/BO switch set to BUCK (the shared power stage is Figure 2.1). The import module places the 12V bench PSU on Port A and the 10V bus on Port B, sourcing current into the bus. The export module places the 10V bus on Port A and a resistor bank on Port B, stepping the voltage down across the resistor to dissipate energy. Current cannot be pushed into the PSU or bank, so the minimum reference current for both SMPS is 0. This satisfies the Port A above Port B rule.

The duty cycle is inverted for these SMPS compared to the PV and supercapacitor boost SMPS due to the inversion of the gate driver. Current sensing is performed by the INA219 driver across the 0.1 ohm Port B shunt. The choice of a dummy resistor is in accordance with the project brief: using a resistor sized to sink the maximum export current instead of driving current into a bench PSU, which would collapse the terminal voltage.

The maximum export power is the sum of maximum PV power and capacitor power. The demo-day configuration of two parallel PV panels supplies a maximum of roughly 8W at the panel side, and the capacitor supplies a maximum of about 13W at peak 0.6A discharging current near the top of its window, making the maximum export power under 20W. Given that the buck SMPS has 10V at the input and the output voltage must be below that, the load bank needs a current of $\frac{P}{V} = \frac{20}{10} = 2$ A and a resistance of $\frac{V}{I} = \frac{10}{2} = 5$ ohm. The dummy load is therefore a 5 ohm bank and the inductor-current reference is capped at 2 A, the point at which the 10V and 2A limits coincide at the 20W corner, and which sits under the module's 2.8A firmware overcurrent trip.

5.1.2 Droop Setpoints: 9.9 V Import and 10.1 V Export

Each module runs a cascaded controller: an outer bus-voltage PI produces a reference current and an inner current PI produces a PWM which changes the voltage (Figure 5.1). The two modules differ only in the sign of the voltage error. The import module computes $v_{\text{err}} = V_{\text{target}} - V_{\text{bus}}$, so it sources more current the further the bus falls below its 9.9V setpoint. The export module computes $v_{\text{err}} = V_{\text{bus}} - V_{\text{target}}$, the sign flipped, so it dissipates more the further the bus rises above its 10.1V setpoint. Each reference current is limited to 2A on both modules (approximately 20W at the bus), the export value set by the worst-case surplus analysis in Section 5.1.1 and the import value by the grid interface rating.

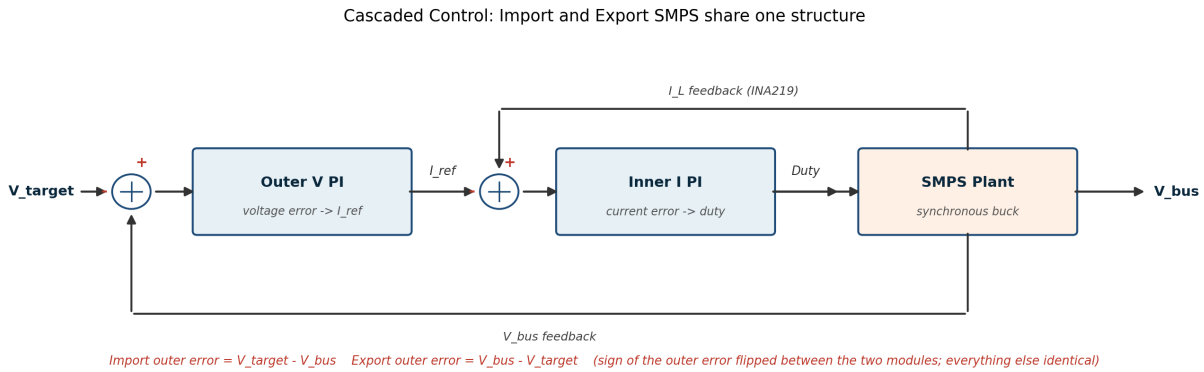


Figure 5.1: Cascaded control structure shared by Import and Export SMPS modules.

The dead-band that the symmetric setpoints create is analysed in Section 2.4. For implementation, a setpoint of 0.1V or below, or a disable command, is treated as fully off, which is how the bus changes from entirely import to entirely export.

5.2 Testing & Implementation

Both bus-facing modules received the same three-layer noise reduction as PV and supercapacitor. The INA219 averaging change is shared through the common `INA219` class in `common.py`: raising the CONFIG word from 0x199F (12-bit, single-sample, 532 us conversion) to 0x1E67 (12-bit, 16-sample averaging, 8.51 ms conversion) inside the constructor covers both modules at once. The four-fold ADC oversample on V_{PSU}, V_{bus} and V_{dummy} and the EMA on V and I traces with $\alpha = 0.05$ are duplicated into each module's main loop. Three-second block averages of P_{import} and $P_{\text{dissipated}}$ are computed inside firmware with reset triggers on V_{target}, I_{max} and enable state, matching the per-module pattern elsewhere; the block-averaged values are exposed in each module's telemetry dictionary but their persistence to the backend SQLite store and surfacing on the dashboard are pending wiring work, so the bench characterisation in Sections 5.3.1 and 5.3.2 reads the steady-state powers from the live filtered telemetry rather than the unwired block-average field.

5.2.1 Anti-Windup and Soft-Start Grace

The outer voltage PI on each module uses clamping anti-windup with a tightly bounded integrator. Left at the default integrator bound the term could wind up much larger than the value that saturates the current reference, taking several seconds to unwind once the bus returns to target. This was observed as the rail voltage fluctuating between 8V and 12V at about 1Hz when the PV and export SMPS ran together without the grid SMPS. We therefore bound the integrator to the range that just saturates the output.

A soft-start grace prevents the undervolt trip. When the system starts up with just the grid module, the bus voltage begins near 0V, which would cause an undervolt trip immediately. The undervolt trip is disabled for a 500ms grace period on every transition from inactive to active (on enable, watchdog clear and trip reset) while overvolt and overcurrent trips stay enabled. The export module carries the same grace logic, but its output current is clamped to 0 when the bus voltage is below target.

5.3 Evaluation & Optimisation

The Export and Import SMPS share a buck topology and a bench characterisation methodology but play opposite roles in the dual-meter scoring framework of Section 2.3. Export sits between

the bus and the dummy resistor: the lecturer's bus meter credits the pre-loss bus value while the dashboard reports the post-loss dissipation, so the dispatch algorithm under-reports export revenue by the conversion loss and operates on slightly off-optimum numbers (Section 6.3.2 covers the same effect on the LED path; the proper fix in both cases is a current sensor between the bus and the SMPS). Maximising η_{export} is the available mitigation. Import is the inverse: the PSU reading is the billed quantity, so every 1% of conversion loss inflates the effective import price per delivered bus Wh directly. The two sweeps are presented in that order.

5.3.1 Measured Export Efficiency

The export SMPS efficiency is $\eta_{\text{export}} = \frac{P_{\text{dissipated}}}{P_{\text{bus,drawn}}}$. The numerator $P_{\text{dissipated}}$ is the firmware's post-loss field, measured at the dummy resistor by the Port B INA219. The denominator $P_{\text{bus,drawn}}$ is not natively instrumented, so a bench DMM is placed in series with the bus-to-export wire and V_{bus} is read off the dashboard. At each operating point the three-second-averaged $P_{\text{dissipated}}$ is recorded against $V_{\text{bus}} \cdot I_{\text{DMM}}$ and the ratio taken.

V_{bus} (V)	Duty D	η at 0.1 A	η at 0.2 A	η at 0.3 A
9.0				
9.5				
10.0				
10.5				
11.0				

Table 5.1: Measured Export SMPS efficiency by bus voltage and load current.

5.3.2 Measured Import Efficiency

The import SMPS efficiency is $\eta_{\text{import}} = \frac{P_{\text{bus,delivered}}}{P_{\text{PSU,drawn}}}$. The pre-loss draw $P_{\text{PSU,drawn}} = V_{\text{PSU}} \cdot I_{\text{PSU}}$ is not measured on the import module because the shunt sits on Port B. Unlike the PV case in Section 3.3, I_{PSU} can be read from the bench PSU's built-in current display with manufacturer-stated accuracy (the same display the lecturer reads on demo day).

Section 2.3 established that any watt lost in the import path is a watt of grid energy missing from the bus, hurting the economic score. Import efficiency therefore gets more attention. The bus-voltage axis is swept across nine points from 7V to 11V in roughly 0.5V steps, wider than the 9.9V to 10.1V dead-band so duty-cycle dependence is captured across the full operating range. Currents of 0.1, 0.2 and 0.3A are used. At each of the 27 operating points the three-second-averaged $P_{\text{bus,delivered}}$ is recorded against $V_{\text{PSU}} \cdot I_{\text{PSU}}$ and the ratio taken.

V_{bus} (V)	Duty D	η at 0.1 A	η at 0.2 A	η at 0.3 A
7.0	0.58	77.8%	86.9%	89.3%
7.5	0.62	79.0%	87.3%	90.0%
8.1	0.67	81.1%	88.1%	90.4%
8.5	0.71	81.7%	88.4%	90.8%
9.0	0.75	82.1%	88.9%	91.2%
9.5	0.79	82.9%	89.6%	91.7%
10.0	0.83	83.8%	89.9%	92.5%
10.5	0.87	84.2%	90.6%	92.7%
11.0	0.92	84.7%	91.3%	93.6%

Table 5.2: Measured Import SMPS efficiency by bus voltage and load current.

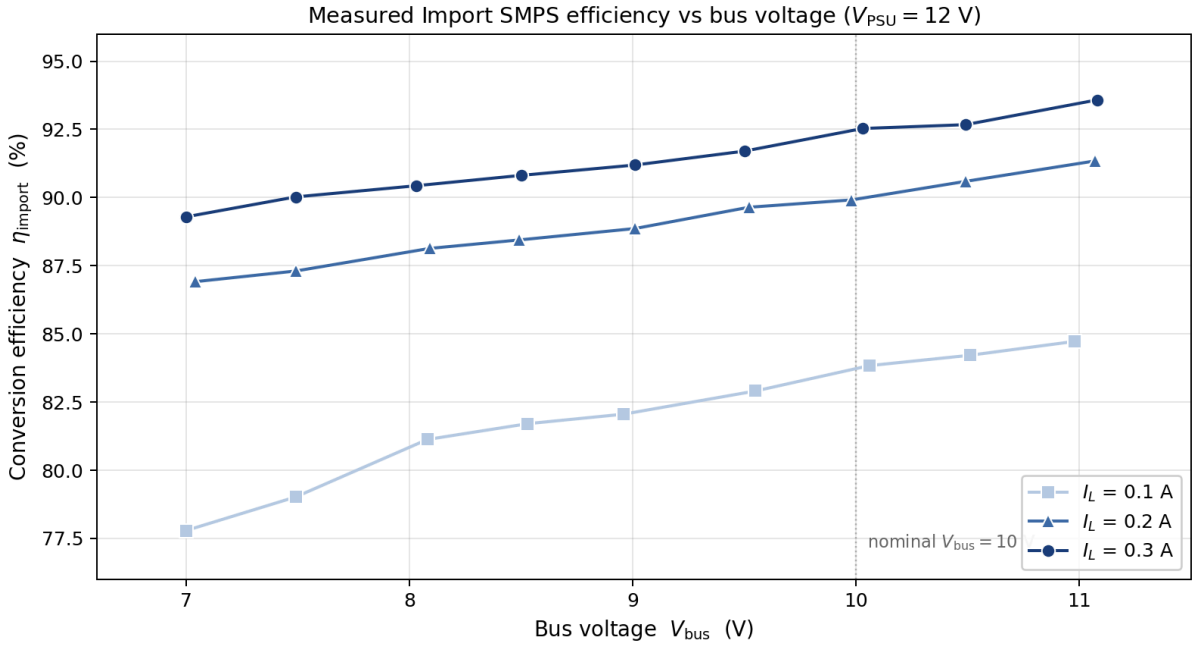


Figure 5.2: Measured Import SMPS efficiency versus bus voltage at three load currents.

At the nominal $V_{\text{bus}} = 10$V the converter delivers 83.8%, 89.9% and 92.5% at 0.1, 0.2 and 0.3A. Both trends (monotonic rise with V_{bus} as duty approaches unity, and monotonic rise with I_L as fixed switching overheads amortise over more delivered power) match the buck loss model of Section 2.3 and confirm the converter is operating away from dropout. In cost terms, the 89.9% at the nominal 10V and 0.2A operating point inflates the effective import price by $\frac{1}{0.899} \approx 1.11$, an 11% premium absorbed by conversion loss that the dispatch algorithm of Section 9 sees directly through its PSU-referenced cost formula.

6 LED Load SMPS

6.1 Design & Simulation

6.1.1 Topology

The LED load is a triple driver assembly: three independent buck channels (red, yellow, green), each a cut-down SMPS driving a 3W, 1A LED, all under one Pico W. Buck is the natural choice because the 3V LED forward voltage is well below the 10V bus. The hardware differs from the bidirectional modules in two ways the firmware must respect. Current is sensed by an external SPI ADC (MCP3208 family) reading three current channels and three voltage channels through one chip select, with a 0.33 ohm shunt rather than 0.10 ohm, so the inductor current is $i_{\mathrm{L}} = 3 \cdot V_{\mathrm{shunt}}$. Each channel has a dedicated enable pin that must be driven high on every tick; the gate driver gates the PWM off whenever its enable is low. The LED forward voltage is reconstructed from the two ADC reads as $v_{\mathrm{LED}} = 2 \cdot v_{\mathrm{v,pin}} - v_{\mathrm{i,pin}}$, correcting for the sense divider and shunt drop. One of the three identical channels is shown in Figure 6.1.

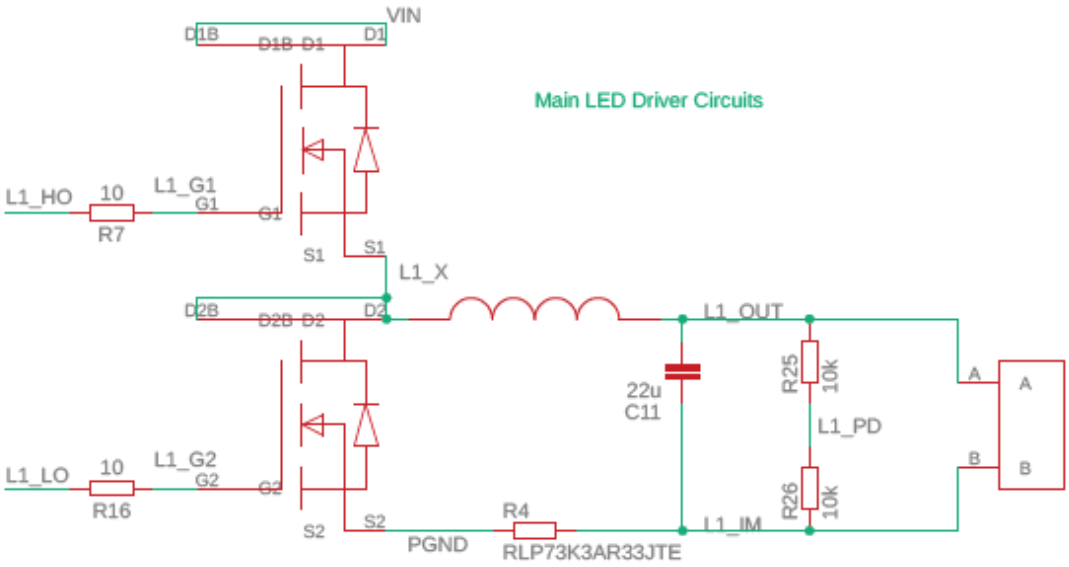


Figure 6.1: LED buck driver power stage, one of three identical channels [8].

6.1.2 Control Mode: CC vs Commanded Power

An LED behaves as a fixed-voltage device, so its power is set by controlling current. The web server hands the system a power figure, so each channel runs commanded-power mode through constant current: power is divided by measured forward voltage to give a current reference $i_{\mathrm{ref}} = \frac{P_{\mathrm{demand}}}{v_{\mathrm{LED}}}$, which a per-channel current PI tracks. The reference is clamped to channel ratings of 3.0W and 1.05A, and a forward-voltage floor guards the division when a string is dark. When demanded power is zero the enable pin is dropped and the PI is reset to stop the integrator winding up while the channel is off.

6.2 Testing & Implementation

The example driver firmware was reworked to drop its external PID dependency in favour of the same inline PI used across the bidirectional modules, giving one consistent controller idiom throughout the project. The per-channel gains and duty saturation range differ from the

bidirectional modules to suit the 0.33 ohm shunt and the driver's own PWM limits. Safety is per channel: if any channel's current exceeds 1.4 times its rated maximum the firmware latches a trip and records which colour tripped, so a faulty string is diagnosable from telemetry instead of showing up as a generic fault.

Two bring-up aids proved useful. A boot-time diagnostic samples every SPI ADC channel several times with no PWM commanded, so a phantom overcurrent from a floating ADC input, a wrong port, or a brown-out can be told apart from a real LED overcurrent by reading the printed values. On shutdown the enables drop first, which gates the PWM off at the driver regardless of commanded duty, before the duties themselves are zeroed, so the strings go dark without residual. A shared 10-second watchdog hook exists for forcing all channels off if commands stop arriving but is currently held inactive (Section 8.3); the operator-driven recovery path is the dashboard Stop action, which broadcasts zero per-channel power through the command proxy.

6.3 Evaluation & Optimisation

Two axes are reported: per-channel power-following accuracy (which verifies Requirement 1) and the reason conversion efficiency is not separately characterised.

6.3.1 Power-Following Accuracy

A 60-second scripted test exercises each channel individually and then together, with commanded and measured per-channel powers logged from the dashboard at 20 Hz. Figure 6.2 overlays the two: the faint trace is commanded power, the dark trace is measured power delivered to the LED string. The sequence is red at 2W for 6s, off, yellow at 2W for 6s, off, green at 2W for 6s, off, all three together at 1, 2 and 3W for 10s, off, and a 15-second simultaneous ramp from 0 to 3W.

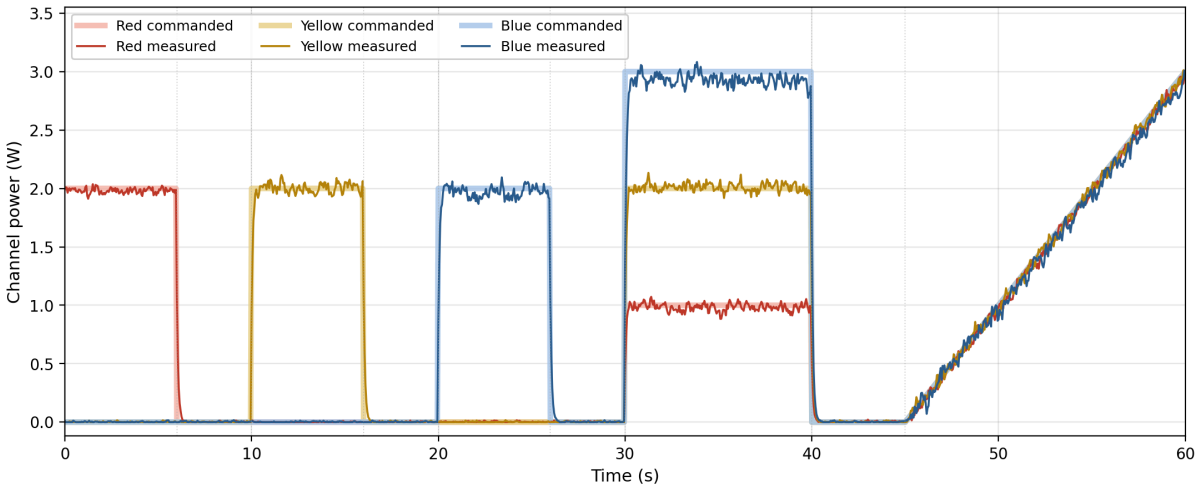


Figure 6.2: Per-channel commanded and measured LED power across the 60-second tracking sequence.

Each channel tracks within about ±3% across all operating points, with the residual driven by per-string forward-voltage variation that the $i_{\text{ref}} = \frac{P_{\text{demand}}}{v_{\text{LED}}}$ division (Section 6.1.2) cannot compensate when v_{LED} is itself estimated through the SPI ADC chain. Step settling on each on/off transition resolves inside one 50 ms dashboard sample. The all-on and ramp phases confirm channels do not interact through the shared bus or SPI ADC, so a multi-colour demand is served additively.

6.3.2 Bus-Side Measurement Gap

The LED Load conversion path sits between the bus meter and the LED string, in the same topological position as Export. The lecturer's bus meter registers $P_{\text{LED,bus}} = P_{\text{LED,channel}} + P_{\text{losses}}$ as load, while firmware reports only the per-channel post-loss values p_{red}, p_{yel}, p_{grn} measured at the LED string. The dispatch algorithm therefore reads an LED demand under-reported by exactly the LED Load buck conversion loss, and the same effect applies on Export where the post-loss dissipation under-reports the bus-credited revenue (Section 5.3). The buck loss could in principle be attributed to either side of the converter, but with no instrumentation between the bus and the SMPS we cannot measure where it actually lands. Maximising η_{LED} is the available lever: at $\eta_{\text{LED}} = 1$ the post-loss value would equal the bus value and dispatch decisions would align exactly with the lecturer's reading; at realistic efficiencies the algorithm runs on slightly optimistic numbers and PnL is marginally worse than a perfectly-informed policy would achieve. The clean fix is a dedicated INA219 between the bus and the SMPS, feeding the true bus-side current to the dashboard so dispatch operates on the lecturer's view of the load; this was identified but not implemented within the project window.

7 Telemetry Filtering

Noise on the per-module power telemetry was bad enough at first bring-up to obscure the operating-point comparisons Sections 3.3, 4.3 and 5.3 depend on. Typical jitter on P_{panel} was several hundred milliwatts peak-to-peak against a steady-state mean of a few watts, with similar relative magnitudes on the other modules. The fix is a three-layer attenuation applied identically across the PV, supercapacitor, import and export modules; it is documented once here and referenced by each per-module section.

7.1 Hardware Averaging

The bidirectional template inherits an INA219 CONFIG register of 0x199F (12-bit, single-sample, 532 us conversion [9]), which lets much of the switching ripple through unaveraged on the shunt channel where it is largest. Reprogramming to 0x1E67 (16-sample averaging, 8.51 ms conversion) across all four bidirectional modules removes the dominant current-channel noise; conversion time rises but stays an order of magnitude below the slowest downstream consumer (the supercapacitor cycle test's 200 ms OCV averaging window, Section 4.3), so the latency is invisible. PV and supercapacitor pick this up through identical edits to their inline `ina_init` functions, and import and export through a single edit to the shared `INA219` class.

Voltage channels are read by the Pico's onboard 12-bit ADC through a resistive divider and walk several LSBs sample-to-sample regardless of INA219 settings. Each `read_va` and `read_vb` call performs four single reads and right-shifts the sum by two to take the mean:

$$V_{\text{oversampled}} = \frac{V[0] + V[1] + V[2] + V[3]}{4}$$

Cost per call is about 10 us, well inside the 1 ms control tick budget. Peak-to-peak jitter falls by roughly 1.6×, with diminishing returns past four samples.

7.2 Exponential Moving Average

A first-order EMA is applied in software on top of the hardware-averaged readings:

$$V_{\text{filt}}[n] = (1 - \alpha) \cdot V_{\text{filt}}[n-1] + \alpha \cdot V[n]$$

with $\alpha = 0.05$ at the 1 kHz sample rate, giving a time constant $\tau \approx 20$ ms and a corner near 8 Hz. The filtered signals drive the dashboard display, the three-second block averages, the P&O power comparison (Section 3.2) and the cycle-test energy integrators (Section 4.3). $\alpha = 0.05$ is bounded both ways: too small (longer τ) and the filter adds phase lag that slows displayed transients on dashboard reads; too large (shorter τ) and it passes switching-ripple aliasing through. 20 ms gives about 60 ms to 95% step settling, which fits inside the 150 ms second-half averaging window of the P&O dwell and the 200 ms tail of the cycle-test settle phases. Figure 7.1 shows the response: a corner near 8 Hz that rejects aliased ripple, with a step settling inside 60 ms so the displayed transient is barely delayed.

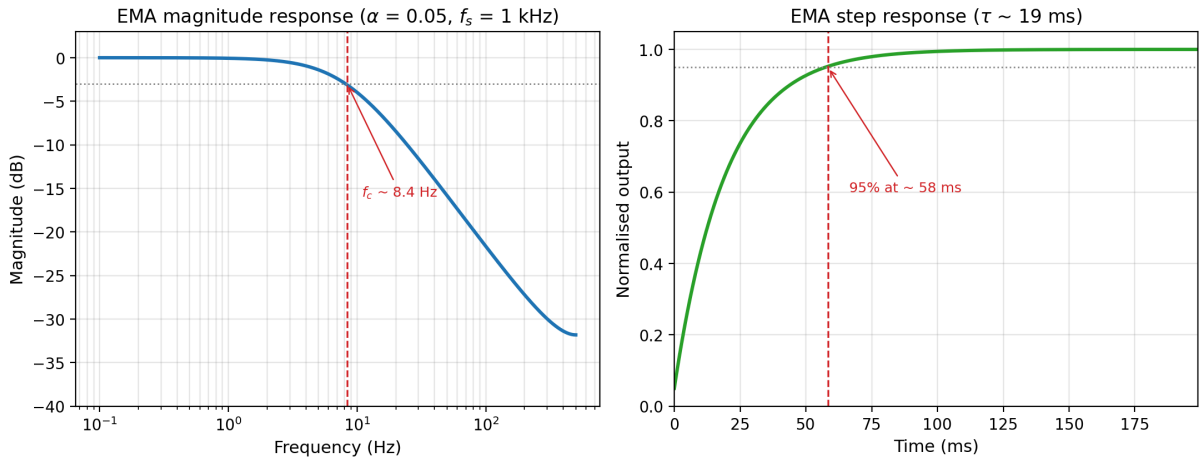


Figure 7.1: Telemetry EMA magnitude and step response at alpha = 0.05.

The inner PI loops and safety guards consume the raw single-sample readings, not the filtered values, for two reasons. First, adding 20 ms of phase lag to a current PI tuned against unfiltered inputs would shift its stability margin and force a retune. Second, the trip thresholds (panel undervolt, bus overvolt, overcurrent) were designed for instantaneous response; a filtered safety guard would react 20 ms late, allowing more energy into a fault before tripping. Keeping the filter on the display and characterisation paths only preserves control-loop behaviour and trip response while still giving the operator clean numbers.

7.3 Three-Second Block Average

Each module computes a true block average of its bus-side power over a three-second window in firmware:

$$P_{\text{avg3s}} = \left(\frac{1}{N}\right) \sum_{i=1}^{N} P[i]$$

with $N = 3000$ at 1 kHz. The window restarts on changes to the commanded operating point (MPPT mode for PV, I_{cmd} for supercapacitor, V_{target} and I_{max} for import and export), so each completed average reflects one operating point. Float-rounding the operating-point tuple before comparison avoids spurious resets from JSON-parser dust. The computed average lives in each module's telemetry dictionary; persistence into the backend SQLite store and rendering on the operator dashboard are scheduled wiring work, so the side-by-side comparisons in Sections 3.3 and 5.3 read the filtered live telemetry directly. Inside the supercapacitor cycle test the block average sits on top of phase changes as a real-time sanity check that the commanded current has settled.

7.4 Evaluation

The three-layer treatment is evaluated by the noise reduction it delivers on the per-module bus-side power telemetry. Figure 7.2 overlays a raw single-sample trace against the same operating point after the three-layer filter, shifted to a 2W PV operating point for visual context.

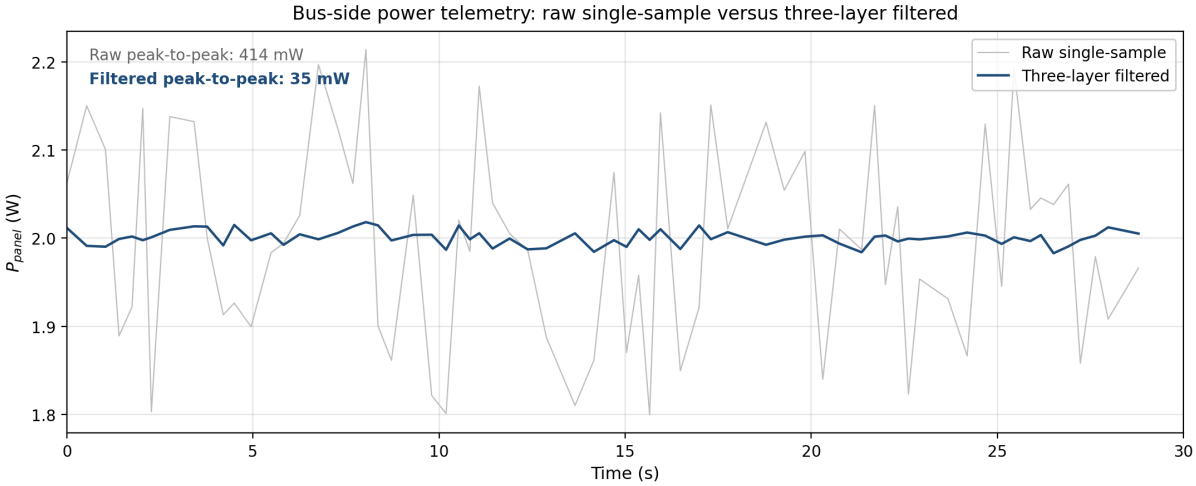


Figure 7.2: Bus-side power telemetry: raw single-sample versus three-layer filtered.

Raw P_{panel} jitter falls from about 300 mW peak-to-peak at the 1 kHz tick to under 10 mW per P&O dwell-averaged comparison (Section 3.2), roughly a 30× reduction. Equivalent attenuation on the supercapacitor, import and export modules brings the three-second-averaged readings inside a few milliwatts of run-to-run reproducibility, fine enough to resolve the 1.4 percentage-point irradiance-to-irradiance spread in the PV efficiency sweep (Section 3.3) and the 1 to 2 percentage-point steps between adjacent supercapacitor charge currents in the round-trip sweep (Section 4.3).

8 Software System

The software system provides the communication, control and monitoring infrastructure for the smart-grid platform. It links the Azure simulation server, backend services, embedded modules and operator dashboard into a unified system.

8.1 Architecture

The lowest tier consists of the firmware running on each Pico-controlled module. Every module hosts a lightweight HTTP server on the Pico's second core, exposing a JSON command endpoint for remote control. Keeping the networking stack on core 1 isolates it from the 1 kHz control loop running on core 0, preventing network activity from introducing jitter into the timing-critical control and safety logic. The same networking thread also pushes a telemetry snapshot to the backend once per second.

The middle tier is a Python FastAPI backend backed by a SQLite database. It stores grid snapshots, telemetry records and algorithm outputs, and acts as the communication bridge between the frontend and the embedded modules. A standalone poller retrieves simulation data from the Azure server and records one deduplicated snapshot per simulation tick.

The top tier is a React single-page application built with Vite. The frontend provides monitoring, visualisation and operator controls through a browser interface.

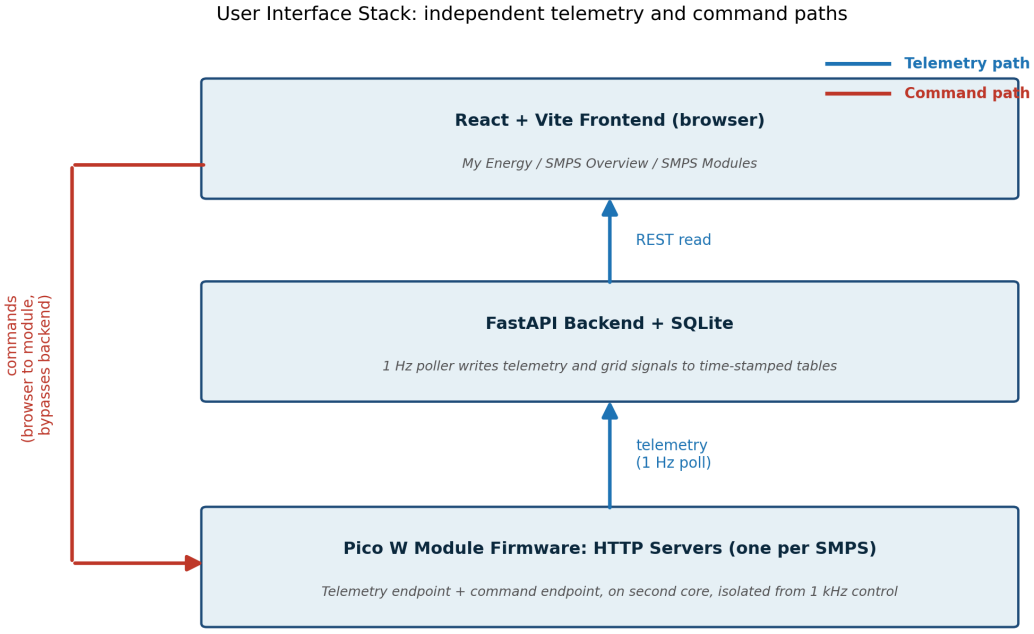


Figure 8.1: Software stack showing telemetry flowing upward to FastAPI and control commands flowing downward through the same backend.

The overall architecture is summarised in Figure 8.1. Telemetry is pushed from the Pico modules to the backend, while both operator commands and EMS-generated commands are routed through the backend before being forwarded to the corresponding module.

8.2 Backend and Control Layer

The backend acts as the central coordination layer for the system. In addition to storing telemetry and simulation data, it hosts the Energy Management System (EMS), which evaluates

the latest grid state together with module telemetry and generates control setpoints for the supercapacitor, load emulator and grid interfaces.

A standalone poller checks the Azure simulation endpoints once per second. When a new `(day, tick)` pair is detected, the corresponding irradiance, pricing and deferrable-load information is retrieved and stored as a single grid-state snapshot. This provides a persistent historical record that can be replayed and analysed independently of the external simulation server.

Multiple algorithms can be registered and selected at runtime through the dashboard, enabling direct comparison between baseline and improved control strategies without requiring firmware changes. Algorithm decisions are logged alongside telemetry and grid-state snapshots, allowing the behaviour of different control policies to be reconstructed and evaluated offline.

The backend also provides run-management facilities, including automatic EMS execution, CSV export of recorded data and a run-scoped profit-and-loss calculation based on imported and exported energy. Together, these features support both real-time operation and post-experiment analysis.

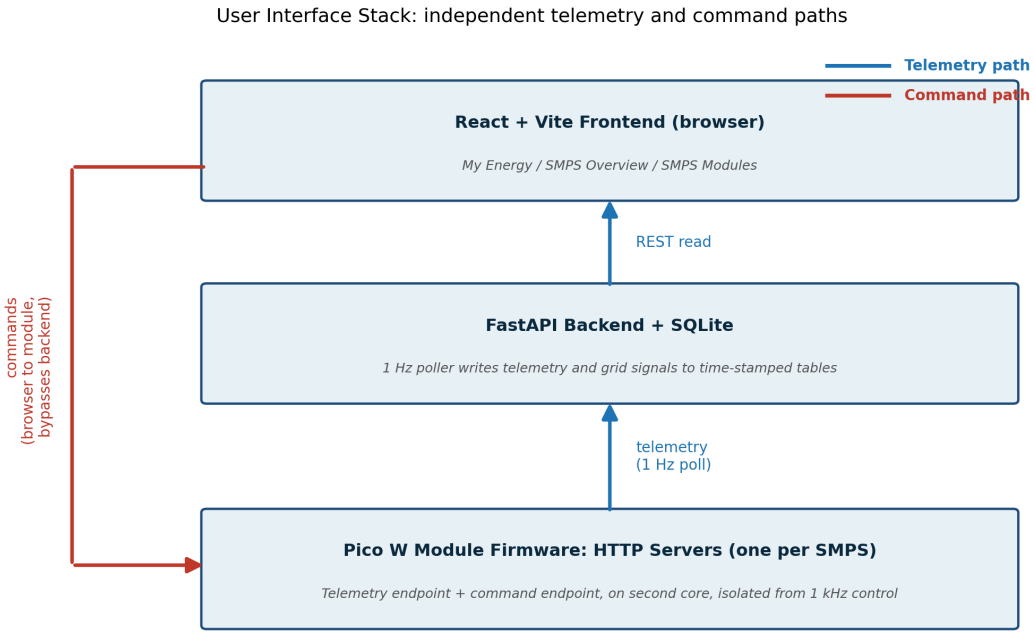


Figure 8.2: EMS CONTROL FLOW

Azure Tick ↓ Grid State ↓ Load Latest Telemetry ↓ Selected Algorithm ↓ Generate Setpoints ↓ Command Dispatcher ↓ Pico Modules

8.3 Frontend Dashboard

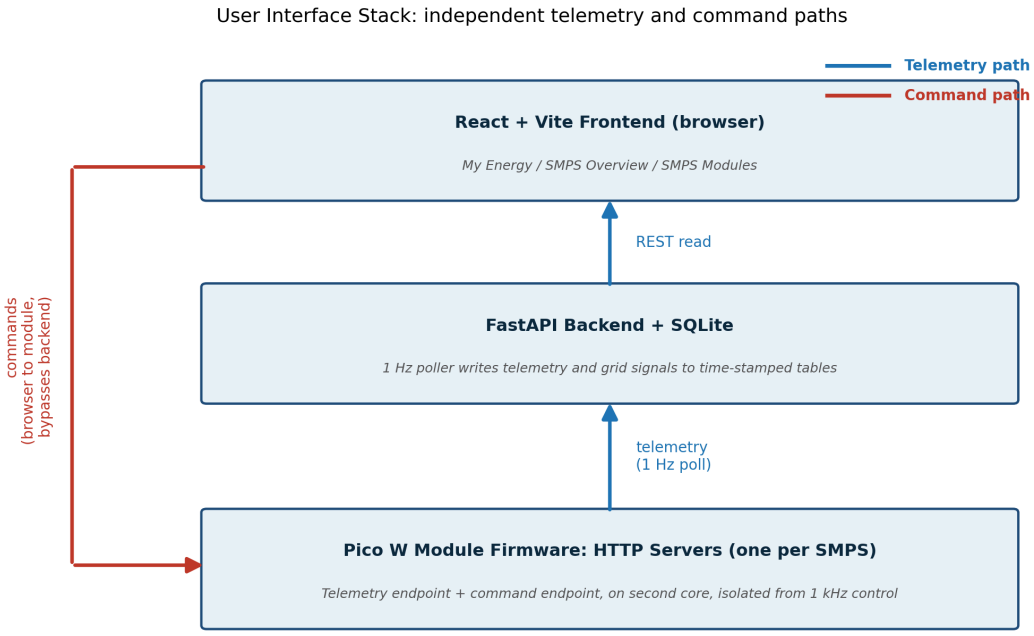


Figure 8.3: HOW TO PRESENT DASHBOARD?

Three operator views are provided. The default **My Energy** dashboard presents a high-level overview of the microgrid, including current demand, solar irradiance, energy prices, active deferrable loads and module health indicators. Historical demand and irradiance data are displayed as charts, while a status panel highlights whether each module is online, stale, tripped, disabled or in a watchdog state.

The **SMPS Overview** page focuses on system-level monitoring. Historical telemetry from the PV, grid, export, supercapacitor and LED modules is visualised using time-series plots of voltages, currents and power flows. This view is intended for observing overall energy movement through the system and diagnosing interactions between subsystems.

The **SMPS Modules** page provides detailed engineering controls. Each module is represented by a dedicated card showing live telemetry, recent history and module-specific controls. Operators can configure MPPT modes on the PV converter, adjust bus-voltage targets for grid and export modules, command supercapacitor charge or discharge currents, set LED load power levels and reset protection trips. The same interface also provides algorithm selection, automatic EMS execution controls, run-scoped profit-and-loss summaries and data export tools for post-run analysis.

8.4 Design Trade-offs

Several implementation choices were revised during integration to improve timing reliability. The first version of the Azure poller attempted to fetch the full grid state at a fixed five-second interval. In practice, a complete cycle of `/demand`, `/sun`, `/price` and `/deferables` requests could take longer than the Azure update period, causing occasional missed ticks. The final design instead checks only `/demand` once per second and treats `(day, tick)` as the change detector. The remaining endpoints are fetched only when a new tick is observed, and are requested in parallel. This keeps tick-detection latency below one second while avoiding unnecessary repeated requests to the slower auxiliary endpoints.

Telemetry collection was changed for a similar reason. Early versions used backend polling to retrieve telemetry from each module. In practice, modules may be temporarily offline during startup, reconfiguration or network interruptions, causing polling requests to block until timeout. Achieving higher telemetry update rates would therefore require either aggressive polling, increasing network traffic, or accepting delays caused by repeated timeout handling. The final implementation uses push-based telemetry: each module sends its latest snapshot to the backend once per second whenever it is online. This makes new measurements available immediately after acquisition, avoids backend resources being tied up waiting for unresponsive modules, and scales more efficiently as the number of modules increases. The trade-off is that telemetry collection now depends on each module maintaining its own network connection and upload loop.

Control commands were kept backend-mediated rather than being sent directly from the frontend to the Pico modules. This allows the EMS algorithm to run headlessly on the backend and issue commands without requiring the dashboard to remain open. It also provides a single control interface for both automated EMS actions and manual operator commands. Centralising command dispatch in the backend ensures that all control actions are logged alongside telemetry and grid-state data, enabling consistent run replay, debugging and post-experiment analysis. In addition, module addressing and communication logic are implemented in one location rather than being duplicated across multiple frontend views or external clients.

8.5 On-Module Fallback Dashboard [R WE MENTIONING THIS AS /TLM IS REMOVED]

As a final redundancy, a single-file dashboard is bundled into the firmware filesystem and served by the module's own HTTP server at its root path. This gives a usable monitoring and control surface with no backend, database or Node toolchain, useful during bench bring-up and as a fallback if the main stack is unavailable on demo day.

9 Algorithm Software

The per-module loops of Sections 3 to 6 keep the bus stable; this section covers the coordinator that decides what each module should be doing. Two policies share the same framework: a reactive naive baseline that the brief asks the smart policy to beat, and a forecast-aware Smart V1 that schedules deferrable loads into cheap ticks and arbitrages the supercapacitor against the evening price spike. Both are evaluated on the same grid under the same scripted day, and the smart policy delivers a 20% net-cost saving averaged across two lab runs.

9.1 Coordinator Loop and Economic Objective

The coordinator is a 1 Hz thread inside the FastAPI backend. Each second it reads the latest grid snapshot and the latest PV and supercapacitor telemetry from the database, evaluates the active algorithm's `decide(state)` function, and dispatches the resulting commands to the four actuators (LED Load, supercapacitor, grid, export) in that fixed order. The loop deduplicates against the Azure tick so the same tick is never applied twice, skips any tick where the grid snapshot is older than 15 seconds, and refuses to dispatch to a module currently latched in a trip. Because each module's inner loop holds the last commanded setpoint autonomously, a coordinator stall cannot destabilise the bus: droop, PI control and per-module clamps run independently at 1 kHz.

The objective is to minimise the net cost of grid energy over a daily Azure cycle while always meeting demand. Under the dual-meter framework of Section 2.3, import is debited at the PSU reading and export credited at the bus reading, so the per-day cost is

$$C = \sum_t \big(P_{\text{PSU}}(t) \cdot p_{\text{sell}}(t) - P_{\text{export,bus}}(t) \cdot p_{\text{buy}}(t)\big) \cdot \Delta t$$

with $P_{\text{PSU}}(t) = P_{\text{import,bus}} \frac{t}{\eta_{\text{import}}}(V_{\text{bus}}, I_L)$ coupling the import draw to the efficiency map of Section 5.3.2. Prices follow the grid's perspective: p_{sell} is what the utility charges us on imports, p_{buy} what it credits us on exports, with $p_{\text{buy}} \approx 0.5 \cdot p_{\text{sell}}$ at every tick. That spread is the headline arbitrage opportunity. The coordinator's levers are deferrable scheduling, supercapacitor charge or discharge, and net import or export, subject to instantaneous demand being met, each deferrable receiving its full energy inside its window, and the supercapacitor and grid modules staying inside their hardware envelopes.

9.2 Algorithm Framework

Each policy exposes a pure `decide(state: GridState) -> Commands` function and is registered with an in-process registry; the operator switches the active policy at runtime from the dashboard and the choice is persisted to disk. `GridState` carries the Azure tick, base demand, PV panel power, bus voltage from the supercapacitor module, supercapacitor terminal voltage, the live current-tick prices and the full deferrable list. `Commands` carries the four actuator payloads plus a diagnostics block (regime, PV derate, effective load, net imbalance, supercapacitor headroom, per-deferrable rates) that the dashboard renders for whichever policy is active. Adding a policy is a register call plus one module.

One detail makes the live path produce the right answer: the Azure feed republishes each deferrable's full original energy at every tick of its window, so a policy that treats `d.energy` as remaining joules would re-allocate the same load over and over. A small served-tracker module accumulates the bus-side energy actually dispatched per deferrable, credits only after

a successful LED command (a dropped command does not bank served energy), and subtracts the running total from `d.energy` before the policy sees it. Without the tracker the LED was over-driven by 233 J on a 615 J target, a 38% over-dissipation; with it, the commanded LED energy lands within 7 J of target on both naive runs and within 41 J on the smart runs, where the residual falls out as export (the LED clamp at 9 W spills surplus into the bus).

9.3 Naive Baseline

The naive policy is a five-step reactive symmetric-droop rule. PV is first derated to bus side by a conservative $\eta_{\mathrm{PV}} = 0.90$ (the measured value in Section 3.3 is 96 to 98%, so the bias keeps the algorithm on the surplus side of modelling error). Active deferrables are then spread linearly across their remaining real-time window so the per-tick rate is $\frac{E_{\text{remaining}}}{5 \cdot N_{\text{ticks remaining}}}$ watts, with adaptive catch-up baked in: under-serving last tick raises the next tick's rate. Effective load is base demand plus the sum of those rates and is commanded to the LED Load, split three ways across red, yellow and green channels with each firmware-clamped to 3 W. The net imbalance $P_{\mathrm{net}} = P_{\mathrm{PV,bus}} - P_{\mathrm{load,eff}}$ is then offered to the supercapacitor through a 1 W deadband (noise floor of the bus), with charge current $\frac{P_{\mathrm{net}}}{V_{\mathrm{bus}}}$ clamped to 0.6 A and refused above 15.7 V, discharge current $|P_{\mathrm{net}}| \frac{}{V_{\mathrm{bus}} \cdot \eta_{\mathrm{cap}}}$ refused below 10.5 V and linearly tapered through the 10.5 to 11 V firmware soft region. Anything the supercapacitor cannot absorb spills to droop: the grid and export modules sit permanently enabled at 9.9 V and 10.1 V respectively and passively make up the residual.

Figure 9.1 shows the naive policy across one Azure day on the grid. The cap power panel is small and twitchy, charging only when noise pushes the bus above the deadband; the grid panel does almost all of the work, importing through the morning, evening and night and exporting only the small midday surplus that the cap could not absorb.

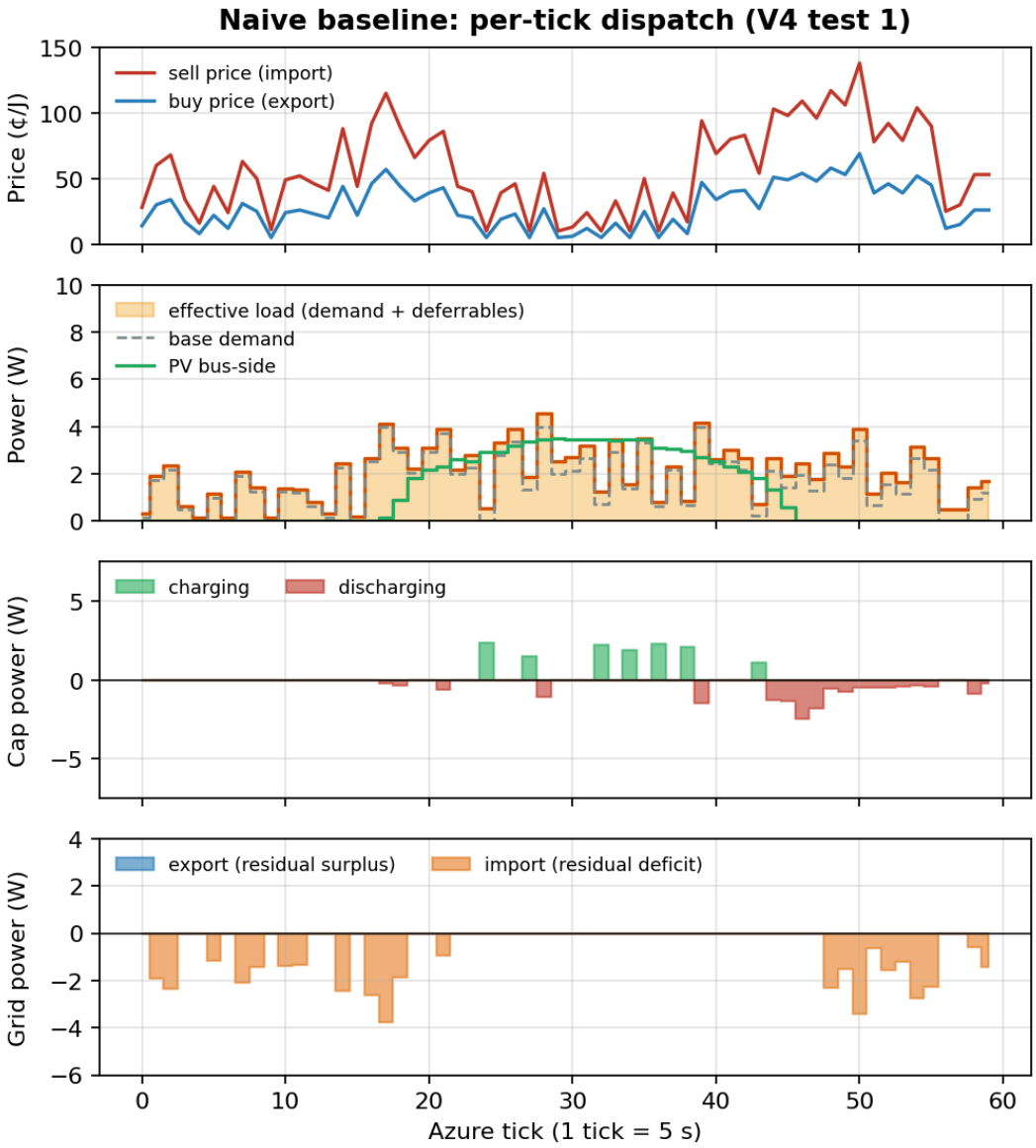


Figure 9.1: Naive baseline: per-tick dispatch over one Azure day (V4 test 1).

9.4 Smart V1: Forecast-Aware Scheduling and Arbitrage

Smart V1 keeps every safety guard of the naive policy (LED clamp, supercapacitor taper, 0.6 A clamp, deadband, discharge-side η_{cap} compensation) and replaces two of its reactive steps with forecast-aware ones. The forecast itself is closed-form: the Azure simulation has a deterministic mean (sun follows a sine over ticks 15 to 45, base demand follows a piecewise-linear envelope, sell price = max(10, 10 + base − sun)), with Gaussian noise on top. Smart V1 evaluates that mean per tick rather than fitting historical data.

The first replacement is the deferrable scheduler. Instead of smearing each deferrable across its window, future ticks are ranked by marginal cost-to-serve: a tick in forecast PV surplus costs p_{buy} (the foregone export revenue), a tick in deficit costs p_{sell} (the imported energy). The scheduler greedy-fills the cheapest ticks first, respecting the LED's 9 W per-tick capacity and giving the tightest-deadline deferrable first pick. The second replacement is the supercapacitor dispatcher. At every tick it compares the current sell price against the best leakage-discounted future sell price across the rest of the day, using a piecewise leakage model fitted to Section

4.3.4 (a 12% dielectric step over the first minute, then 0.25% per Azure tick). It emits one of six actions: end-of-day drain past tick 55, arbitrage discharge when now beats the best future by more than 20 ¢/J, arbitrage charge when forecast yield clears charging cost by the same margin, an empty-cap refill rule that lowers the import bar when a forecast peak is within eight ticks, and a fall-back passive rule that mirrors the naive deadband when no arbitrage triggers fire. Two pre-peak guards (refuse passive discharge below 70 ¢/J after tick 42, and a fixed price floor instead of forecast comparison) were added after a lab trace showed the supercapacitor being drained at moderate prices and arriving empty at the real evening peak.

Figure 9.2 shows the smart policy on the same day. The supercapacitor panel now has a large midday charge burst (ticks 24 to 32, where PV surplus and cheap prices coincide) and an evening discharge burst (ticks 45 to 50, where the sell price clears 100 ¢/J). The grid panel imports most of its energy in the cheap morning and exports the midday surplus that exceeds the LED clamp.

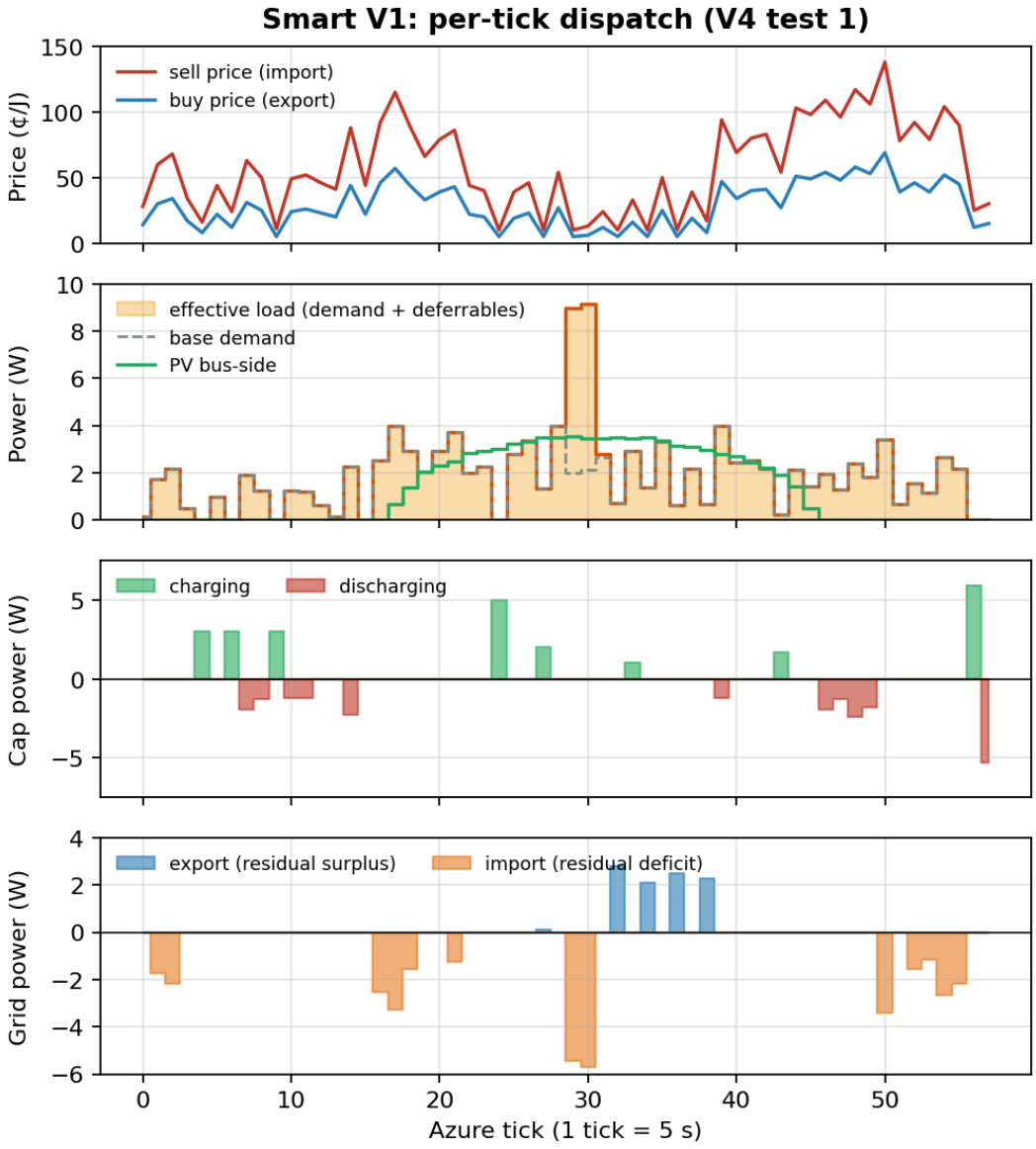


Figure 9.2: Smart V1: per-tick dispatch over the same Azure day (V4 test 1).

9.5 Evaluation

Each policy was run on the grid against the same fixed price, demand and irradiance profile (514 J base demand, 100.6 J of deferrables, 615 J LED target). Net cost integrates the grid module's `p_import` and the export module's `p_dissip` trapezoidally over every consecutive sample pair, multiplied by the sell or buy price in effect at the earlier timestamp; sample gaps longer than 30 s are skipped. The same integrator drives the `/algo/run/cost` endpoint the dashboard renders during a run, so the numbers in Table 9.1 are directly comparable to a live P&L card.

Run	Import (¢)	Export (¢)	Net (¢)	Saving vs naive
Naive, test 1	34,789	70	34,719	baseline
Naive, test 2	33,493	63	33,430	baseline
Smart, test 1	27,863	450	27,413	21.0%
Smart, test 2	27,447	384	27,064	19.0%

Table 9.1: V4 lab cost summary (cents). Smart V1 saves 20% on average.

Figure 9.3 attributes the saving to where each policy spent its import allowance. Smart V1 imports roughly 3.6 times as much energy in the cheapest band (below 30 ¢/J) and correspondingly less in the 30 to 100 ¢/J middle bands, which is where the naive policy is forced to draw whenever the bus sags. Total imported energy is almost the same across both policies (476 J for smart versus 463 J for naive, averaged across the two paired tests): the saving comes from when the energy is bought, not how much. The supercapacitor cycles about 2.2 times as hard under smart (120 J charged, 98 J discharged on average, versus 54 J and 49 J under naive) and the grid exports about five times as much (36 J versus 7 J on average), the export being PV surplus that exceeds the 9 W LED clamp on peak midday ticks while the supercapacitor is also fully charging. The dominant lever is therefore temporal arbitrage on the supercapacitor: cheap-band imports park energy in the cap, which is then discharged later to displace what would otherwise be 70 to 100 ¢/J imports.

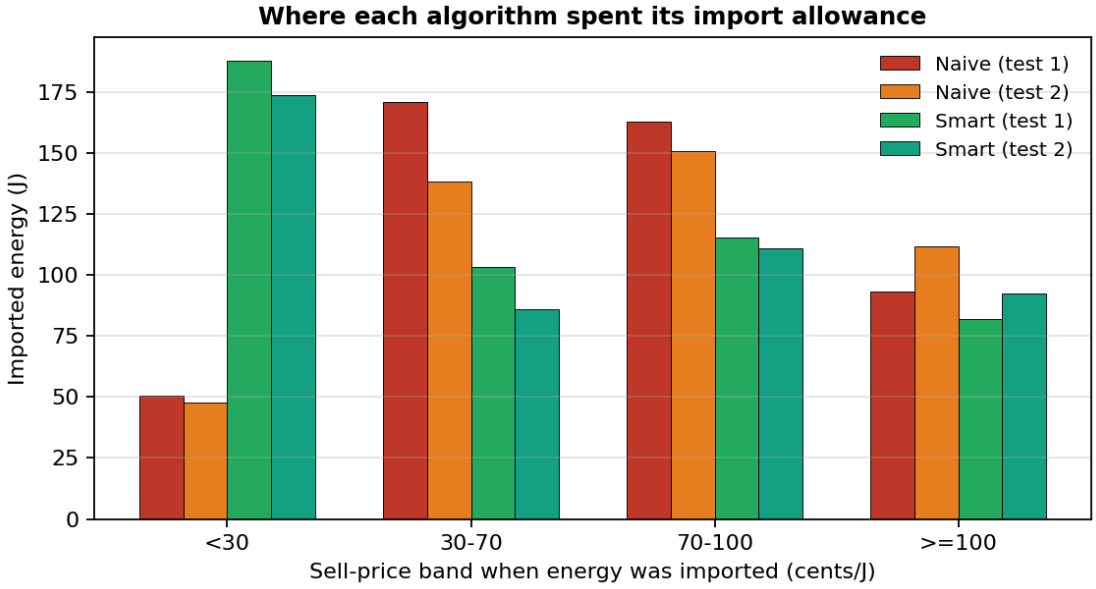


Figure 9.3: Imported energy by sell-price band: smart shifts imports into the cheap band.

9.6 Auxiliary Power Overhead

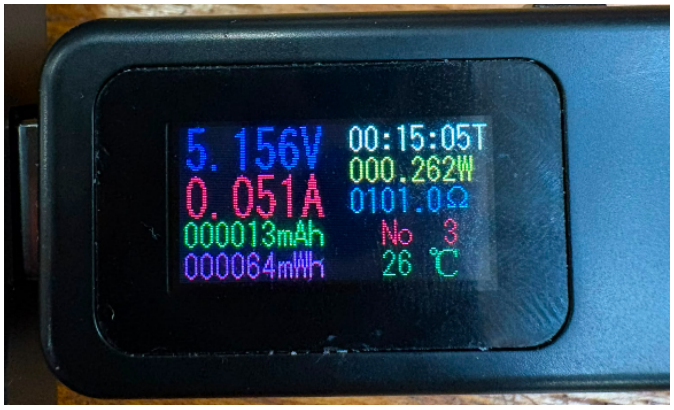


Figure 9.4: USB inline meter: a single Pico W draws 0.262 W on its 5 V rail.

The five Pico W controllers run from a separate USB powered hub, unmetered by the algorithm. A USB inline meter reads 0.262 W per module (Figure 9.4), so the aggregate fixed draw is 1.31 W or 393 J per 300 s Azure day. Total energy throughput across the grid (PV panel-side, supercapacitor charge and discharge, grid import and export, LED dissipation) averaged 1724 J per day, or 5.75 W. The controllers therefore account for $\frac{1.31}{1.31+5.75} \approx 19\%$ of total grid power. However, we note that this is a fixed cost: with a small-scale grid it is highly inefficient, but with a larger supercapacitor, higher demand, it can improve the efficiency.

10 Mechanical Design & Project Polish

10.1 3D-Modelled SMPS Enclosure

The four bidirectional SMPS boards sit in a 3D-printed chassis that protects the exposed power electronics from accidental shorts and keeps the assembly stable enough to move and demo without disturbing the wiring.

The v1 print (Figure 10.1, left) had solid side walls and a solid base. It was mechanically fine but trapped heat: with no intake from the sides or below, the lower boards had no convection path.

The v2 revision (Figure 10.1, right) added vertical slots in the side walls and vents in the base, so cool air enters from below and rises through to the open top. 3 mm standoff ribs on the inside walls hold each board off the wall so air flows between them.

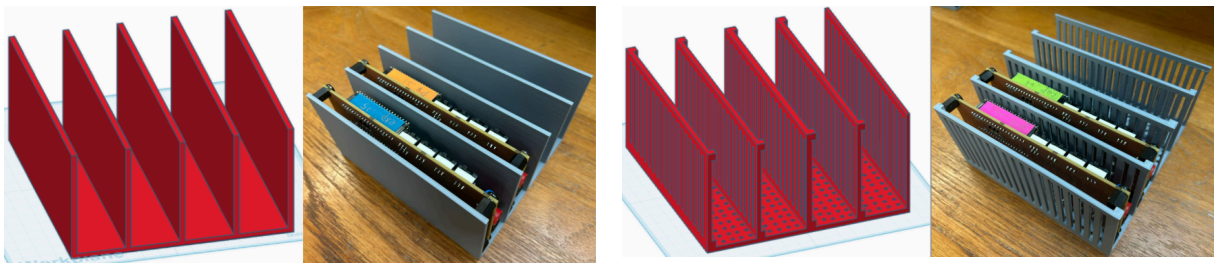


Figure 10.1: SMPS stand iteration: v1 (left) and v2 (right)

11 System-Level Testing & Evaluation

11.1 Test Scenarios

System-level evaluation has two test categories: per-module SMPS efficiency characterisations (the headline numbers reported here) and full demo-day economic-dispatch dry runs against scripted price-and-irradiance profiles. The first is described in this section; the second is introduced in Section 9.

11.1.1 Per-Module Efficiency Sweeps

Each SMPS module is characterised across a grid of operating points, with the test method matched to the instrumentation available on its port assignment. The methods are summarised below; full details are in the per-module evaluation subsections.

PV SMPS (Section 3.3): manual measurement using a DMM in series with the PV-to-bus wire. The pre-loss panel power $P_{\text{panel,in}}$ is from firmware (both factors are Port B); the post-loss bus power $P_{\text{bus,out}}$ needs the external DMM since Port A is uninstrumented. Both axes are complete: twelve bus-voltage points from 7.1 to 12.55V at each of four irradiance settings (25%, 50%, 75%, 100%), 48 measurements total. The four curves give means of 96.5%, 96.7%, 97.0% and 97.9% rising monotonically with irradiance, flat with bus voltage to within ±0.3%.

Supercapacitor SMPS (Section 4.3): fully automated cycle test from the host dashboard. The test integrates bus-side E_{in} and E_{out} at the 1 kHz tick and reports $\eta_{\text{RT}} = \frac{E_{\text{out}}}{E_{\text{in}}}$ from a single charge-and-discharge cycle. The sweep covers six currents (0.1 to 0.6A in 0.1A steps) at a 10V bus, five bus voltages (8.0 to 10.0V) at 0.2A, and six hold durations (0 to 300 s) at 0.2A. Measured round-trip efficiency is about 88% with no hold, falling to 64% after a 300 s hold at peak charge.

Import SMPS (Section 5.3.2): manual measurement using the bench PSU's built-in current readout for the pre-loss $P_{\text{PSU,drawn}}$ side; the post-loss $P_{\text{bus,delivered}}$ comes from firmware. Bus voltage is swept across nine points from 7V to 11V in 0.5V steps; current is set at 0.1, 0.2 and 0.3A by the outer voltage PI. The 27-point grid gives η_{import} rising monotonically from 77.8% (low current, low bus) to 93.6% (high current, high bus), with 83.8%, 89.9% and 92.5% at the nominal $V_{\text{bus}} = 10$V at 0.1, 0.2 and 0.3A.

Export SMPS (Section 5.3.1): mirrors the import procedure but with the DMM in series on the bus-to-export wire (the firmware shunt sits on the resistor side), so $P_{\text{dissipated}}$ is from firmware and $P_{\text{bus,drawn}} = V_{\text{bus}} \cdot I_{\text{DMM}}$. The sweep grid and headline numbers are in Section 5.3.1. Under the dual-meter framework (Section 2.3) this efficiency governs the dashboard-vs-meter gap rather than entering the cost formula directly: the bus meter credits the bus-side export at p_{buy}, while the dashboard's post-loss telemetry under-reports that revenue by the conversion loss.

11.1.2 Communication & Integration Tests

Azure Tick Acquisition Telemetry Collection End-to-End Command Path [Azure Update->CMD Sent->Execution] Module Disconnect & Recovery?

11.2 Performance Metrics

The headline metrics from the efficiency sweeps are tabulated below. Each entry is a ratio of post-loss output power to pre-loss input power, averaged over the relevant operating window. For the supercapacitor, only the round-trip efficiency is reported per operating point because

the per-phase split is confounded by charge redistribution (Section 4.3). For PV and import, a single number is reported per operating point. All numbers are accompanied by their operating point (bus voltage, current, irradiance) and the raw P_{in} and P_{out} readings so the ratios can be audited.

Two derived system-level checks are also reported. First, the steady-state bus voltage. At idle, the bus holds within about ±0.05V of the 10V nominal, well inside the ±0.1V droop deadband from Section 2.3, which is sized as a safety margin against transient sag rather than for steady-state regulation. Second, the energy balance closure across the bus for full-cycle scenarios:

$$E_{\text{PV,bus}} + E_{\text{import,bus}} = E_{\text{export,bus}} + E_{\text{LED,bus}} + E_{\text{supercap,bus}} + E_{\text{losses,bus}}$$

The unmodelled $E_{\text{losses,bus}}$ residual captures bus-wire I^2R and any uncharacterised dissipation. It should be small (a few % at most) if the per-module post-loss measurements are consistent. A large residual on a closure check is treated as a regression signal for the per-module efficiency numbers.

11.3 Requirements Verification Matrix

Each of the brief's six requirements is mapped below to the subsystem that realises it, the test that verifies it, and its current status. Requirement 5 (the economic dispatch algorithm) is the main item still in progress (Section 9); the rest are implemented and verified by the tests in the cited sections.

Req	Requirement	Realised by	Verification	Status
1	Supply domestic LEDs per web demand	LED Load SMPS (Section 6)	Power-tracking test (Section 6.3)	Implemented; tracking within ±3% across channels
2	Extract PV energy with MPPT	PV SMPS (Section 3)	MPPT mode comparison; efficiency sweep	Implemented; P&O shipped as default with Fixed and LUT fallbacks
3	Store excess in supercapacitor, no batteries	Supercapacitor SMPS (Section 4)	Cycle test; round-trip efficiency	Implemented
4	Import and export to grid, metered	Import/Export SMPS (Section 5)	Import efficiency sweep; export efficiency sweep; bus dead-band	Implemented; export sweep scheduled
5	Minimise cost; schedule storage and deferrables; beat naive	Algorithm (Section 9)	Dry run versus naive baseline	In progress
6	UI for current and historic energy data	User Interface (Section 8)	Live dashboard; telemetry plots	Implemented

Table 11.1: Brief requirements: verification method and current status.

12 Conclusion

The project delivered a five-module DC microgrid regulating a shared 10V bus across PV generation, supercapacitor storage, a bidirectional grid interface and a programmable LED load. The two grid-facing converters coordinate the bus through symmetric droop with a 0.2V deadband, holding the bus inside ±0.05V of nominal without a central coordinator and degrading safely under communication loss. The PV module reaches 96 to 98% conversion across irradiance and bus voltage under selectable lookup-table and P&O MPPT. The import converter spans 78 to 94% across its operating sweep. The supercapacitor is a command-driven store with a self-contained cycle test that returns about 88% round-trip efficiency from bus-side energy alone. The LED load follows commanded power through constant current with per-channel tracking inside ±3%. A three-layer telemetry filter chain reduced raw bus-side power jitter by roughly 30×, and an in-house React and FastAPI interface provides monitoring and control on independent paths so a backend outage does not block operator action. The main remaining work before demo day on 18 June 2026 is the economic dispatch algorithm that turns the price and demand signals into cost-minimising decisions.

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